

# Partial Results on the WOWII-61 Graffiti Conjecture

Haoyu Chen\*

22 August 2026

## Abstract

Conjecture 61 of the *Written on the Wall II* collection of computer-generated graph conjectures asserts that  $\text{residue}(G) + \lceil \text{diam}(G)/3 \rceil \leq f(G)$  for every connected graph  $G$  on at least two vertices, where  $\text{residue}(G)$  is the number of entries left standing when the Havel–Hakimi reduction is run to completion on the degree sequence of  $G$ , and  $f(G)$  is the order of a largest induced forest. *The conjecture is not resolved here.* This paper records what an automated campaign proved about it, with the epistemic status of each statement stated explicitly and separately.

The inequality survives 150 905 graph evaluations, among them every connected graph on at most eight vertices, with no violation. It holds unconditionally when  $\text{diam}(G) \in \{1, 2, 3, 5, 6, 9\}$ , and whenever the decycling number of  $G$  is small against its matching number; two conditional certificates cover 98.8% of connected graphs on at most seven vertices and 99.2% of a sample on eight, and every graph they miss in that range has diameter exactly 4. At diameter 4 the conjecture is equivalent to a dichotomy:  $f(G) \geq \alpha(G) + 2$  or  $\text{residue}(G) \leq \alpha(G) - 1$ . Most of the paper proves cases of that dichotomy. Writing  $\tau = n - \alpha$  for the size of a minimum vertex cover and  $B$  for its complement in  $V$ , we settle  $\tau \leq 3$  completely; and, under the standing hypothesis  $\text{residue}(G) = \alpha(G)$  that a counterexample would have to satisfy, we prove a  $\tau$ -uniform budget inequality and a one-unit strengthening of it, a master budget inequality for the low-degree vertices of  $B$ , a rigidity theorem which pins the  $B$ -universal layer of the residual configuration space to an explicitly described family  $\text{GFan}(\tau, L, \nu)$ , and the elimination of the sub-family  $\text{Fan}(\tau, L) = \text{GFan}(\tau, L, 1)$  for every  $L \geq 2$ . Together these force any surviving configuration to have  $\tau \geq 4$  and at least three vertices of  $B$  of degree at most  $\tau$ .

Statements are reported in three separated tiers according to how far they have passed the campaign’s adversarial-review protocol. Sections 4–9 contain only statements that survived two review rounds by judges from distinct model families with no mathematics defect; Section 10 collects a family that has passed one such round; and Section 11 reports a computer-assisted elimination whose current statement has passed none — together with the

---

**\*Disclosure of the production pipeline.** This paper reports the partial results of an autonomous attack on a conjecture that it does *not* resolve. Every mathematical statement below was produced by automated systems: an orchestrating agent and a per-problem owner agent (Claude Fable 5 and Claude Opus 5) that designed the lines of attack, adjudicated the returning reports and drafted this document; adversarial reviewing sessions run by models from deliberately different families (Claude Opus 5; Qwen3.8-Max through its web interface; and OpenAI GPT-5.6 and GPT-5.3 through the `codex` command-line interface, in the configurations named `sol` and `spark` respectively); and machine checkers, which are purpose-written Python enumerators together with independent from-scratch re-implementations used only for adjudication. No human supplied a definition, a lemma, a proof idea, or a correction. Problem selection, task routing, review scheduling and the drafting of this paper were likewise automated. Every numbered *assertion* carries an explicit verification marker in the sense of Section 1.3; definitions and open problems assert nothing and carry none. Two of the campaign’s own statements were refuted by its own later review rounds, and both refutations are reported here rather than quietly repaired (Remark 4.9 and Section 11.2). Responsibility for what follows rests with the reader who checks it.

false quantifier that a review round found in its predecessor, and the counterexample family that refuted it.

## Contents

|           |                                                                        |           |
|-----------|------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                                    | <b>3</b>  |
| 1.1       | The conjecture . . . . .                                               | 3         |
| 1.2       | What this paper contains . . . . .                                     | 4         |
| 1.3       | Verification conventions . . . . .                                     | 4         |
| 1.4       | Relation to the campaign’s other output . . . . .                      | 5         |
| <b>2</b>  | <b>Notation and the basic package</b>                                  | <b>6</b>  |
| 2.1       | The Havel–Hakimi trajectory . . . . .                                  | 6         |
| 2.2       | The cover $B$ and the reductio hypothesis . . . . .                    | 7         |
| <b>3</b>  | <b>Elementary certificates, and the concentration at diameter four</b> | <b>7</b>  |
| 3.1       | Diameter at most three . . . . .                                       | 7         |
| 3.2       | The 2-packing certificate . . . . .                                    | 8         |
| 3.3       | The matching certificate . . . . .                                     | 8         |
| 3.4       | An equivalent form . . . . .                                           | 9         |
| 3.5       | Falsification, and where the residual difficulty sits . . . . .        | 9         |
| 3.6       | The diameter-four dichotomy . . . . .                                  | 10        |
| 3.7       | Named obstacles . . . . .                                              | 10        |
| <b>4</b>  | <b>The Havel–Hakimi toolkit</b>                                        | <b>11</b> |
| <b>5</b>  | <b>Frame lemmas at diameter four</b>                                   | <b>14</b> |
| <b>6</b>  | <b>Small covers: <math>\tau \leq 3</math></b>                          | <b>15</b> |
| <b>7</b>  | <b>A <math>\tau</math>-uniform budget</b>                              | <b>17</b> |
| 7.1       | The bound . . . . .                                                    | 17        |
| 7.2       | The master budget . . . . .                                            | 19        |
| 7.3       | The two rigid low configurations . . . . .                             | 20        |
| <b>8</b>  | <b>The fan family</b>                                                  | <b>21</b> |
| 8.1       | A tail lemma . . . . .                                                 | 24        |
| <b>9</b>  | <b>Rigidity of the <math>B</math>-universal layer</b>                  | <b>25</b> |
| <b>10</b> | <b>Statements at one adversarial review round</b>                      | <b>26</b> |
| <b>11</b> | <b>A computer-assisted elimination, and a false quantifier</b>         | <b>29</b> |
| 11.1      | The statement, and the case list . . . . .                             | 29        |
| 11.2      | The refutation . . . . .                                               | 30        |

|                                                                                        |           |
|----------------------------------------------------------------------------------------|-----------|
| <b>12 What remains</b>                                                                 | <b>32</b> |
| 12.1 The residual configuration space . . . . .                                        | 32        |
| 12.2 Two combinatorial conjectures that would close the $B$ -universal layer . . . . . | 32        |
| 12.3 The one review round that would close the certification surface . . . . .         | 33        |
| <b>13 Limitations</b>                                                                  | <b>33</b> |
| <b>A Certification ledger</b>                                                          | <b>34</b> |
| <b>B Verification-artifact inventory</b>                                               | <b>36</b> |
| <b>C The <math>\nu \leq 6</math> enumeration, in full</b>                              | <b>40</b> |

# 1 Introduction

## 1.1 The conjecture

For a finite simple graph  $G$  on vertex set  $V$ ,  $|V| = n$ , write  $d_1 \geq d_2 \geq \dots \geq d_n$  for the degree sequence in non-increasing order. The *Havel–Hakimi reduction* deletes the first entry  $d_1$  and subtracts 1 from each of the next  $d_1$  entries, re-sorts, and repeats; it halts when the largest remaining entry is 0. The number of entries standing at that moment is the *residue*  $\text{residue}(G)$ . Let  $f(G)$  denote the order of a largest induced forest of  $G$ ,  $\alpha(G)$  its independence number,  $\mu(G)$  its maximum matching size,  $\nabla(G) := n - f(G)$  its decycling (feedback vertex) number,  $\Delta(G)$  its maximum degree,  $m$  its number of edges, and  $\text{diam}(G)$  its diameter.

**Conjecture 1.1** (WOWII-61). For every connected graph  $G$  on  $n \geq 2$  vertices,

$$\text{residue}(G) + \left\lceil \frac{\text{diam}(G)}{3} \right\rceil \leq f(G). \quad (\text{C61})$$

We take the conjecture in the form in which it is recorded in the `formal-conjectures` corpus [1], in the file `FormalConjectures/WrittenOnTheWallIII/GraphConjecture61.lean`, whose statement is

```
theorem conjecture61 (G : SimpleGraph a) [DecidableRel G.Adj] (h : G.Connected) :
  (residue G : R) + ceil ((G.diam : R) / 3) <= (G.largestInducedForestSize : R)
```

with `[Fintype a]` `[DecidableEq a]` `[Nontrivial a]`. That file attributes the statement to the *Written on the Wall II* collection of conjectures generated by the `Graffiti.pc` program; we cite the formal file rather than the original collection because the formal file is the object our campaign actually consumed, and because we did not consult the original source (Section 13). Since `G.diam` is a natural number and both sides of the Lean statement are casts of naturals, the goal is equivalent to the integer inequality (C61) with  $\lceil d/3 \rceil$  the ordinary integer ceiling.

Three definitional points matter and are worth stating, because a formalized conjecture can be proved in a sense its author did not intend. First, `residue` in the corpus is `residueAux` applied to the descending degree sequence, where `residueAux [] = 0`, `residueAux (0 :: s) = 1 + |s|`, and otherwise the function recurses on one Havel–Hakimi step; the Python transcription used throughout the campaign reproduces this literally, including `List.splitAt` truncation and `N-truncated subtraction`. Second,  $f(G)$  is  $\max\{|S| : G[S] \text{ acyclic}\}$ . Third, since  $G$  is connected and the vertex type finite, `G.diam` is the ordinary diameter. As a sanity check on the transcription,

| statements                                                                                                                        | section | tier                                         |
|-----------------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------|
| Lemma 2.1, Fact 2.2, Lemma 3.1, Theorem 3.2, Lemma 3.3, Theorem 3.4, Corollaries 3.5–3.6, Theorem 3.7, Corollary 3.8, Theorem 3.9 | 3       | elementary; full proofs given                |
| Lemma 4.1, Lemma 4.2, Lemma 4.3, Lemma 4.4, Lemma 4.5, Theorem 4.6, Corollary 4.7                                                 | 4       | certified (rows R-1 – R-5)                   |
| Lemma 5.1, Observation 5.2, Fact 5.3, Lemma 5.4                                                                                   | 5       | certified (root-reduction block)             |
| Theorem 6.1, Theorem 6.2                                                                                                          | 6       | certified (row R-13)                         |
| Lemma 7.1, Theorem 7.2, Theorem 7.3, Corollary 7.5, Theorem 7.6, Corollaries 7.7–7.8, Propositions 7.9–7.11, Corollary 7.10       | 7       | certified (rows R-16 – R-18)                 |
| the six fan lemmas, Lemma 8.12, Observation 8.5, Theorem 8.9, Corollary 8.11                                                      | 8       | certified (rows R-9, R-10, R-14, R-15, R-19) |
| Lemma 9.1, Corollary 9.2, Theorem 9.3, Corollary 9.5                                                                              | 9       | certified (rows R-20, R-21)                  |
| Lemmas 10.1–10.3, Theorem 10.4, Corollary 10.5, Corollaries 10.6–10.7                                                             | 10      | one round only                               |
| Theorem 11.1, Corollary 11.3                                                                                                      | 11      | no completed round                           |
| Conjectures 12.1 and 12.2                                                                                                         | 12      | conjecture                                   |

Table 1: The statements reported here and their verification tiers. Registry row identifiers are the campaign’s own; the evidence chain for each is reproduced in Appendix A.

$\text{residue}(K_2) = 1$ , matching the `example` in the Lean file, and  $\text{residue}(C_n) = \lceil n/3 \rceil$  for  $n = 3, \dots, 9$ ; both are recomputed by every verification script cited in Appendix B as its first action.

## 1.2 What this paper contains

Nothing here resolves Conjecture 1.1. What the campaign produced is (i) a falsification stage that found no counterexample, (ii) an elementary package that settles six of the possible diameters unconditionally and reduces the measured residual difficulty to diameter 4, (iii) a reformulation of the diameter-4 residue as a dichotomy, and (iv) a long chain of structural results narrowing the configurations in which a counterexample to that dichotomy could live. Table 1 lists the statements that this paper reports and the tier each occupies.

## 1.3 Verification conventions

Because this is an automated campaign’s output, the epistemic status of each statement is recorded explicitly, and it is not uniform. The campaign’s protocol has a stage devoted to adversarial review, internally called S3, which works as follows. A completed argument is submitted — in a fresh session, without the solver’s reasoning trace, and under a read restriction forbidding the reviewer to open any other artifact of the campaign — to a model from a family different from the one that produced it, with named *joints* assigned by the dispatcher, a pre-written attack plan per joint that names the mechanism to test but never the expected answer, a mandatory block of control cases that the accepted machinery must *not* also prove, and instructions to classify every objection as a mathematics defect or a bookkeeping defect and to return a verdict. Each returning report is then adjudicated by the owner agent, which reproduces every claim in it from

scratch with an independent implementation before upholding or rejecting it. A round counts as clean for a statement only if it returns no mathematics defect against that statement, and — a rule enforced strictly, and enforced against this campaign’s own reading in one case reported in Section 11 — only for the text *as reviewed*, so that a round on text that was subsequently repaired does not transfer to the repaired text.

Five markers are used.

- **[certified:** two review rounds, distinct model families, zero mathematics defects; registry rows] — the statement has a complete proof, reproduced here or summarised with an explicit pointer, and it has passed *two* review rounds by judges from distinct model families with zero mathematics defects against it, at least one of the two on the final repaired text. The registry rows named in the marker are the campaign’s own ledger entries; Appendix A reproduces the judge, artifact and date for each. This is the strongest marker used in this paper, and it is a statement about a review process, not about a machine-checked proof: nothing here is formalized.
- **[one round:** round; second family pending] — a complete proof, one clean round, and a second family still owed. The named round is identified in Appendix A.
- **[no completed round:** reason] — a complete proof in the campaign’s judgement, and no completed review round on the current text. Section 11 is the only place this marker appears in a headline statement, and the reason it appears there is that the previous statement at that position was refuted.
- **[elementary;** proof reproduced in full] — an elementary argument, three to fifteen lines, reproduced here in full, so that the reader can discharge it directly without reference to any review process.
- **[computational:** scope] — a completed finite computation of exactly the stated scope. A computation is evidence of the stated scope and of nothing beyond it; in particular a *failed* bounded search is never a proof of impossibility, and where such a search is reported the text says so.

A sixth marker, [conjecture; evidence], marks statements believed but not proved.

Two conventions about numerics are enforced throughout and are worth stating once. First, *numbers before statements*: every numerical block cited below was run before the statement it accompanies was written, and the text says so where it matters. Second, and more important, the *reductio firewall*: almost every statement in Sections 4–11 is proved under the standing hypothesis  $\text{residue}(G) = \alpha(G)$ , and the campaign’s own results say that hypothesis is unsatisfiable in the configurations concerned. Numerical runs on such configurations are therefore *not* instances of those statements; they are mechanism checks on the same combinatorial machinery off the hypothesis, and they corroborate the shape of an argument rather than its conclusion. Where a numerical block is quoted below, it is quoted with that distinction attached.

## 1.4 Relation to the campaign’s other output

The pipeline that produced this paper is the one described in the reports on four OEIS conjectures [7] and on a conjecture of Fernandes [8]; those papers report resolved problems formalized in Lean 4, and this one reports an unresolved problem with no formalization at all. The difference is deliberate and is the point of the tiering above: an automated campaign that only publishes what it finished would be reporting a biased sample of its own behaviour. A companion working

paper on the last open finite implication of the Equational Theories Project [9] takes the same view and uses the same marker discipline, which we have followed here.

We make no claim of priority for any statement below, and no claim that any of them is new. Automated literature search found no prior treatment of Conjecture 1.1 beyond its appearance in the collection and in the formal corpus, but automated searches miss things (Section 13).

## 2 Notation and the basic package

Throughout,  $G$  is a finite simple connected graph on  $n \geq 2$  vertices with vertex set  $V$  and edge set  $E$ ,  $|E| = m$ . We abbreviate  $\alpha = \alpha(G)$ ,  $f = f(G)$ ,  $d = \text{diam}(G)$ ,  $\Delta = \Delta(G)$ ,  $\nabla = \nabla(G) = n - f$ ,  $\mu = \mu(G)$ . For  $S \subseteq V$ ,  $G[S]$  is the induced subgraph,  $N(v)$  the neighbourhood of  $v$  and  $N[v] = N(v) \cup \{v\}$ .

### 2.1 The Havel–Hakimi trajectory

Run the Havel–Hakimi reduction on the degree sequence of  $G$  with vertex labels attached, so that at each step we may speak of *which vertex* is deleted. Let  $s = s(G)$  be the number of steps performed before the process halts, let  $D_1, \dots, D_s$  be the values of the successive deleted entries (the *heads*), let  $K$  be the set of vertices deleted as heads, and call the  $n - s$  vertices never deleted the *survivors*. The *block* of step  $j$ , written  $\text{block}_j$ , is the set of  $D_j$  entries decremented at that step: the prefix of the remaining entries, sorted non-increasingly, after deleting the step- $j$  head.

Ties are not broken canonically by the definition, and the campaign’s protocol treats tie-sensitivity as a first-class failure mode: every numerical run cited below is performed under the canonical sort *and* under several randomised adversarial tie-breaks, and every proof step below that could depend on a tie is either tie-invariant or says which tie-break it fixes and why fixing it is legitimate.

**Lemma 2.1** (residue =  $n - s$ ). [*elementary; proof reproduced in full*] Let  $s(G)$  be the number of Havel–Hakimi steps and  $D_1, \dots, D_s$  the successive head values. Then

- (1)  $\text{residue}(G) = n - s(G)$ ;
- (2)  $D_1 = \Delta \geq D_2 \geq \dots \geq D_s \geq 1$  and  $\sum_{i=1}^s D_i = m$ .

*Proof.* (1) Each step shortens the list by exactly one entry, and the recursion stops on a list whose head — hence, the list being sorted non-increasingly, every entry — is 0, returning its length. After  $s$  steps the length is  $n - s$ .

(2) Havel–Hakimi preserves graphicality [2, 3], so at every step the head  $D_i$  is at most the number of remaining strictly positive entries; the N-truncated subtraction therefore never truncates, and one step decreases the sum of the list by exactly  $2D_i$  ( $D_i$  removed as the head,  $D_i$  further units removed by the decrements). The sum starts at  $2m$  and ends at 0, giving  $\sum_i D_i = m$ . For monotonicity, the new list is obtained from the old one by deleting the head and decreasing some entries, so its maximum cannot increase.  $\square$

**Fact 2.2** (Favaron–Mahéo–Saclé). [*computational: asserted by the verification script on all 30 995 graphs of stage E and never violated; cited, not reproved*]  $\text{residue}(G) \leq \alpha(G)$  for every graph  $G$ .

This is used as a black box throughout. It is due to Favaron, Mahéo and Saclé [4]; a short proof is due to Griggs and Kleitman [5]. The bibliographic data was supplied by an automated literature pass and should be re-checked before any formalization (Section 13); the inequality

itself is asserted by the campaign's verification script on the whole exhaustive range and never fails there.

## 2.2 The cover $B$ and the reductio hypothesis

Fix a *maximum* independent set  $A$  and put  $B := V \setminus A$ ,  $\tau := |B| = n - \alpha$ . Then  $B$  is a minimum vertex cover, every  $a \in A$  has  $N(a) \subseteq B$  and hence  $\deg(a) \leq \tau$ , and every  $b \in B$  has at least one neighbour in  $A$  (otherwise  $A \cup \{b\}$  would be independent). We write  $\deg_A(v) = |N(v) \cap A|$ ,  $\deg_B(v) = |N(v) \cap B|$ ,  $e_B = |E(G[B])|$ , and

$$\nu := \binom{\tau}{2} - e_B$$

for the number of non-adjacent pairs inside  $B$ . Since  $A$  is independent, every edge has an endpoint in  $B$ , so  $m = \sum_{b \in B} \deg(b) - e_B$ .

By Lemma 2.1(1) and Fact 2.2,  $n - s = \text{residue} \leq \alpha = n - \tau$ , so  $s \geq \tau$  always. The campaign's central standing hypothesis is the opposite extreme:

**Definition 2.3** (reductio hypothesis). The *reductio hypothesis* is  $\text{residue}(G) = \alpha(G)$ , equivalently  $s(G) = \tau(G)$ .

Under the reductio hypothesis the Havel–Hakimi process must terminate in exactly  $\tau$  steps, and it is that scarcity of steps which the whole of Sections 4–11 exploits. We also fix the low/high split of  $B$  that organises everything from Section 7 onwards:

$$B_{\text{hi}} := \{b \in B : \deg(b) \geq \tau + 1\}, \quad p := |B_{\text{hi}}|, \quad B_{\text{lo}} := B \setminus B_{\text{hi}}, \quad L := |B_{\text{lo}}| = \tau - p.$$

For  $S \subseteq B$  let  $\nu(S)$  be the number of non-adjacent pairs inside  $S$ , and put  $\bar{m} := \nu(B_{\text{hi}})$ . A vertex  $b \in B$  is *B-universal* if it is adjacent to every other vertex of  $B$ ;  $B_{\text{lo}}^+$  denotes the set of low vertices that are *not*  $B$ -universal.

## 3 Elementary certificates, and the concentration at diameter four

This section is self-contained and elementary. Every proof in it is short and is given in full, so that the reader can discharge the whole section without reference to the review process; only Fact 2.2 and the classical graphicality theorem behind Lemma 2.1 are used as black boxes.

### 3.1 Diameter at most three

**Lemma 3.1.** *[elementary; proof reproduced in full] For every connected  $G$  with  $n \geq 2$ :  $f(G) \geq \alpha(G) + 1$ .*

*Proof.*  $G$  is connected with  $n \geq 2$ , so it has an edge; hence no independent set is all of  $V$ . Take a maximum independent set  $A$  and any  $v \in V \setminus A$ . In  $G[A \cup \{v\}]$  the set  $A$  is independent, so every edge is incident with  $v$ : the graph is the star with centre  $v$  and leaves  $N(v) \cap A$ , together with  $|A| - |N(v) \cap A|$  isolated vertices. A star plus isolated vertices is acyclic, and  $|A \cup \{v\}| = \alpha(G) + 1$ .  $\square$

**Theorem 3.2.** *[elementary; proof reproduced in full] If  $\text{diam}(G) \leq 3$  then (C61) holds.*

*Proof.*  $n \geq 2$  and  $G$  connected give  $d \geq 1$ , so  $\lceil d/3 \rceil = 1$ . By Fact 2.2 and Lemma 3.1,  $\text{residue}(G) + 1 \leq \alpha(G) + 1 \leq f(G)$ .  $\square$

This alone settles 89.7% of connected graphs on at most seven vertices and 78.4% of those on eight.

### 3.2 The 2-packing certificate

Call  $B' \subseteq V$  a *2-packing* if  $\text{dist}(b, b') \geq 3$  for all distinct  $b, b' \in B'$ .

**Lemma 3.3** (star-forest lemma). *[elementary; proof reproduced in full] If  $A$  is independent,  $B'$  is a 2-packing and  $A \cap B' = \emptyset$ , then  $G[A \cup B']$  is a forest — indeed a disjoint union of stars and isolated vertices.*

*Proof.*  $B'$  is independent because its members are at distance  $\geq 3 > 1$ , and  $A$  is independent; hence every edge of  $G[A \cup B']$  joins  $A$  to  $B'$ . If distinct  $b, b' \in B'$  had a common neighbour  $a \in A$  then  $\text{dist}(b, b') \leq 2$ , a contradiction; so the sets  $N(b) \cap A$ ,  $b \in B'$ , are pairwise disjoint. Therefore  $G[A \cup B']$  is the disjoint union of the stars with centre  $b$  and leaf set  $N(b) \cap A$ , together with the uncovered vertices of  $A$ .  $\square$

**Theorem 3.4** (2-packing certificate). *[elementary; proof reproduced in full] Let  $\rho^*(G) := \max\{|B'| : B' \text{ a 2-packing with } B' \cap A = \emptyset \text{ for some maximum independent set } A\}$ . Then  $f(G) \geq \alpha(G) + \rho^*(G)$ ; consequently, if  $\rho^*(G) \geq \lceil d/3 \rceil$  then (C61) holds for  $G$ .*

*Proof.* Lemma 3.3 applied to a witnessing pair  $(A, B')$  gives an induced forest of order  $\alpha + |B'|$ ; then apply Fact 2.2.  $\square$

**Corollary 3.5.** *[elementary; proof reproduced in full] For every connected  $G$  with  $n \geq 2$ :  $f(G) \geq \alpha(G) + \lfloor (d-1)/4 \rfloor + 1$ .*

*Proof.* Fix a maximum independent set  $A$  and a shortest path  $P = v_0 v_1 \cdots v_d$  realising the diameter; a shortest path is induced and  $\text{dist}(v_i, v_j) = |i - j|$ . Since  $A \cap V(P)$  is independent in  $P$ , no two consecutive indices lie in  $A$ ; hence  $I := \{i \in [0, d] : v_i \notin A\}$  meets  $\{i, i+1\}$  for every  $0 \leq i \leq d-1$ . Consequently consecutive elements of  $I$  differ by at most 2,  $\min I \leq 1$  and  $\max I \geq d-1$ .

Greedily set  $i_1 := \min I$  and  $i_{j+1} := \min\{i \in I : i \geq i_j + 3\}$  while such an  $i$  exists. Because consecutive elements of  $I$  differ by at most 2 we have  $i_{j+1} \leq i_j + 4$ , so  $i_k \leq 1 + 4(k-1)$ . The process stops at index  $k$  with  $\max I < i_k + 3$ , i.e.  $i_k > \max I - 3 \geq d-4$ . Combining,  $1 + 4(k-1) > d-4$ , i.e.  $k > (d-1)/4$ , so  $k \geq \lfloor (d-1)/4 \rfloor + 1$ .

The set  $B' := \{v_{i_1}, \dots, v_{i_k}\}$  has pairwise index gaps  $\geq 3$ , hence pairwise distances  $\geq 3$  (equalities hold along a shortest path), and  $B' \cap A = \emptyset$  by construction. Apply Lemma 3.3.  $\square$

**Corollary 3.6.** *[elementary; proof reproduced in full] (C61) holds for every connected  $G$  with  $\text{diam}(G) \in \{1, 2, 3, 5, 6, 9\}$ .*

*Proof.* By Corollary 3.5 and Fact 2.2 it suffices that  $\lfloor (d-1)/4 \rfloor + 1 \geq \lceil d/3 \rceil$ . For  $d \geq 10$  this fails, since  $\lfloor (d-1)/4 \rfloor + 1 \leq (d+3)/4 < d/3 \leq \lceil d/3 \rceil$ , the middle inequality being  $3d+9 < 4d$ . A finite check of  $d = 1, \dots, 9$  leaves exactly  $d \in \{1, 2, 3, 5, 6, 9\}$ .  $\square$

The constant  $1/4$  here, rather than the  $1/3$  the conjecture asks for, is exactly the quantitative gap; see Obstacle O2 in Section 3.7.

### 3.3 The matching certificate

**Theorem 3.7.** *[elementary; proof reproduced in full] If  $\mu(G) \geq \nabla(G) + \lceil d/3 \rceil$  then (C61) holds.*

*Proof.* A maximum matching has  $\mu$  pairwise disjoint edges and an independent set contains at most one endpoint of each, so  $\alpha + \mu \leq n$ . Hence by Fact 2.2,  $\text{residue}(G) \leq \alpha \leq n - \mu \leq n - \nabla - \lceil d/3 \rceil = f(G) - \lceil d/3 \rceil$ .  $\square$

**Corollary 3.8.** *[elementary; proof reproduced in full](C61) holds whenever  $\nabla(G) \leq \lceil d/2 \rceil - \lceil d/3 \rceil$ ; in particular for every forest, and for every graph of decycling number 0.*

*Proof.* A shortest path with  $d$  edges contains a matching of size  $\lceil d/2 \rceil$ , so  $\mu(G) \geq \lceil d/2 \rceil$ ; apply Theorem 3.7. For a forest,  $\nabla = 0 \leq \lceil d/2 \rceil - \lceil d/3 \rceil$ , the right-hand side being non-negative for every  $d$ .  $\square$

### 3.4 An equivalent form

**Theorem 3.9.** *[elementary; proof reproduced in full](C61) is equivalent to*

$$s(G) \geq \nabla(G) + \left\lceil \frac{\text{diam}(G)}{3} \right\rceil,$$

*that is:* the Havel–Hakimi process must run for at least  $\lceil d/3 \rceil$  steps more than the decycling number.

*Proof.* Immediate from  $\text{residue} = n - s$  (Lemma 2.1) and  $f = n - \nabla$ .  $\square$

The classically known part of this is  $s \geq \nabla + 1$ , which is Fact 2.2 combined with Lemma 3.1. So the conjecture asks for a gain of  $\lceil d/3 \rceil - 1$  over the classical bound, and *only* that.

### 3.5 Falsification, and where the residual difficulty sits

*Remark 3.10* (falsification stage). [computational: 150 905 graph evaluations in stages A–D, 0 violations; stage E re-checks 30 995 of them against three further invariants, also 0 violations; `wowii61_verify.py` all, runtime  $\approx 100$  s on one core] No counterexample to (C61) was found. The stages are: (A) every connected graph on  $2 \leq n \leq 7$  vertices, 995 graphs, from the graph atlas; (B) a complete cover of all 11 117 connected graphs on 8 vertices, obtained by enumerating all  $853 \times 127 = 108\,331$  pairs (connected  $H$  on 7 vertices, non-empty  $N(v) \subseteq V(H)$ ) — every connected graph has a non-cut vertex, so every connected 8-vertex graph is such an  $H + v$ , and no isomorphism rejection is needed since duplicates only cost time; (C) 1 209 structured graphs with  $n \leq 26$  (paths, cycles, path and cycle powers  $P_n^k, C_n^k$  for  $k \leq 4$ , lexicographic blow-ups  $P_m[K_r], C_m[K_r], P_m[E_r], C_m[E_r]$ , chains of cliques sharing cut vertices, theta graphs, caterpillars, random trees); (D1) 7 994 uniform random connected graphs on  $n = 9, 10$ ; (D2) 32 376 graphs on  $n = 9, \dots, 13$  produced by simulated annealing that minimises the slack  $f - \text{residue} - \lceil d/3 \rceil$ . Stage E additionally *asserts*  $\text{residue} \leq \alpha$ ,  $f \geq \alpha + 1$ ,  $f \geq \alpha + \lfloor (d-1)/4 \rfloor + 1$  and (C61) itself on every graph it touches, so a counterexample to any of them aborts the run.

The minimum of the slack  $f - \text{residue} - \lceil d/3 \rceil$  over all connected graphs with  $n \leq 8$ , by diameter, is

| $d$                | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|--------------------|---|---|---|---|---|---|---|
| min slack          | 0 | 0 | 0 | 0 | 1 | 2 | 2 |
| $\min(f - \alpha)$ | 1 | 1 | 1 | 1 | 2 | 2 | 4 |

so in that range the inequality is tight exactly up to  $d = 4$  and becomes loose beyond.

*Remark 3.11* (measured coverage). [computational: all 995 connected graphs with  $n \leq 7$  and a 30 000-graph prefix of the  $n = 8$  enumeration] Evaluating the three proved certificates P1 = Theorem 3.2, P2 = Theorem 3.7, P3 = Theorem 3.4 gives coverage 89.7/74.1/96.1% and 98.8% jointly for  $n \leq 7$ , and 78.4/98.0/93.3% and 99.2% jointly at  $n = 8$ . Of the 267 graphs in that range covered by none of the three, *every one has diameter exactly 4*; among them  $f - \alpha = 1$  for 223 and  $= 2$  for 44,  $\alpha - \text{residue} \in \{1, 2\}$ , the slack is 0 for 112 and 1 for 155, and  $\nabla \in \{1, 2, 3\}$ .

We stress the scope of Remark 3.11: it is a statement about  $n \leq 8$ . The diameters left open in general by Corollary 3.6 are 4, 7, 8 and every  $d \geq 10$ ; what the measurement says is that in the exhaustively checked range the residual instances all sit at  $d = 4$ , not that  $d = 4$  is the only residual diameter for all  $n$ . Section 13 returns to this.

### 3.6 The diameter-four dichotomy

At  $d = 4$  we have  $\lceil d/3 \rceil = 2$ , while Corollary 3.5 yields only  $f \geq \alpha + 1$  and Fact 2.2 yields only  $\text{residue} + 2 \leq \alpha + 2$ . The shortfall is exactly one unit, which is the following statement.

**Proposition 3.12** (the  $d = 4$  dichotomy). [*elementary; proof reproduced in full*] Suppose that for every connected  $G$  with  $\text{diam}(G) = 4$ ,

$$f(G) \geq \alpha(G) + 2 \quad \text{or} \quad \text{residue}(G) \leq \alpha(G) - 1. \quad (\text{D4})$$

Then (C61) holds for every connected  $G$  with  $\text{diam}(G) = 4$ , and hence, with Corollary 3.6, for every connected  $G$  with  $\text{diam}(G) \in \{1, 2, 3, 4, 5, 6, 9\}$ .

*Proof.* Let  $\text{diam}(G) = 4$ , so  $\lceil d/3 \rceil = 2$ . In the first case  $\text{residue} + 2 \leq \alpha + 2 \leq f$  by Fact 2.2. In the second,  $\text{residue} + 2 \leq \alpha + 1 \leq f$  by Lemma 3.1.  $\square$

Everything from Section 4 onwards is directed at (D4). Negating it gives the object the campaign calls the hard core, in two tiers which must be kept apart — several statements below hold in the weaker one, and conflating them was itself a defect found in review.

**Definition 3.13** (frame and hard core). The *hard-core frame* is:  $G$  connected, not a forest,  $\text{diam}(G) = 4$ , and  $f(G) = \alpha(G) + 1$ . The *hard core* is the frame together with the *reductio* hypothesis  $\text{residue}(G) = \alpha(G)$  — and, after Theorem 6.2, together with  $\tau \geq 4$ .

(D4) is exactly the assertion that the hard core is empty. A third tier is needed too, and is used for most of Section 4: *reductio only*, meaning the hypothesis  $\text{residue} = \alpha$  with no structural assumption whatever. That tier is not vacuous even where the hard core is:  $C_5$  has degree sequence  $[2, 2, 2, 2, 2]$ ,  $\alpha = 2$ ,  $\tau = 3$  and  $\text{residue} = 2$ , so it satisfies the *reductio* while  $\text{diam} = 2$  and  $f = 4 \neq \alpha + 1$ , i.e. it lies outside the frame entirely.

### 3.7 Named obstacles

Four obstacles were isolated, each with an explicit witness. They explain why the elementary package stops where it does, and they constrain what a proof can look like. All witnesses below are re-verified by `wowii61_verify.py` w.

**O1: the two slacks do not separate.** Put  $X := \alpha - \text{residue} \geq 0$  and  $Y := f - \alpha \geq 1$ ; then (C61) says exactly  $X + Y \geq \lceil d/3 \rceil$ . The tree  $W_2$  on 8 vertices with edges  $\{01, 12, 23, 24, 25, 26, 73\}$  has  $d = 4$ ,  $\alpha = 5$ ,  $\text{residue} = 5$ ,  $f = 8$ , so  $X = 0$  at  $d = 4$ ; the

graph  $W_1$  on 8 vertices with edges  $\{01, 12, 13, 14, 25, 56, 71, 75\}$  has  $d = 4$ ,  $\alpha = 6$ ,  $f = 7$ , residue = 5, so  $Y = 1$  at  $d = 4$  (and  $W_1$  is simultaneously a tight instance of (C61):  $7 = 5 + 2$ ). Hence no proof bounding one of  $X$ ,  $Y$  alone by  $\lceil d/3 \rceil - O(1)$  can work: the diameter information must enter the Havel–Hakimi process and the forest construction simultaneously. In particular the two obvious strengthenings “ $\alpha + 1 \geq \text{residue} + \lceil d/3 \rceil$ ” and “ $f \geq \alpha + \lceil d/3 \rceil$ ” are both false, witnessed by  $W_2$  and  $W_1$  respectively.

**O2: the 1/4 versus 1/3 gap.** Corollary 3.5’s bound  $\rho^* \geq \lfloor (d-1)/4 \rfloor + 1$  is best possible *for that argument*: the extremal configuration has  $A \cap V(P) = \{v_0, v_2, v_4, \dots\}$ , so the path vertices available off  $A$  are the odd-indexed ones and a 2-packing among them must skip every second one. Getting  $d/3$  requires admitting pairs  $v_i, v_{i+2}$  into  $B'$ , which is legitimate exactly when  $v_i$  and  $v_{i+2}$  have at most one common neighbour in  $A$  — i.e. one must control 4-cycles between consecutive odd-indexed path vertices and  $A$ . The smallest graph where that control genuinely fails is  $W_4$  on 6 vertices with edges  $\{02, 13, 24, 25, 34, 35\}$ :  $d = 4$ ,  $\alpha = 4$ ,  $f = 5$ , residue = 3, slack 0, and  $\rho^* = 1 < 2 = \lceil d/3 \rceil$ .

**O3: residue admits no usable vertex-deletion recursion.** Both natural recursions fail. For  $v$  of minimum degree, residue( $G$ )  $\leq 1 + \text{residue}(G - N[v])$  is false: on  $W_5$ ,  $n = 6$ , edges  $\{02, 05, 13, 14, 23, 34, 35\}$ , degrees  $[2, 2, 2, 4, 2, 2]$ ,  $v = 0$ , one has residue( $G$ ) = 3 but residue( $G - N[0]$ ) = 1. For  $v_0$  a diametral endpoint, residue( $G$ )  $\leq \text{residue}(G - N[v_0])$  fails — for *every* diametral endpoint — in 36.4% of the 7228 connected graphs with  $n \leq 8$  and  $d \geq 4$ . The structural reason is common to both: deleting  $N[v]$  lowers  $n$ , which lowers residue, and lowers the degrees of the survivors, which raises it, and the two effects are not comparable.

**O4: degree-sequence-only bounds are exactly one short at  $d = 4$ .** For  $W_1$  ( $n = 8$ ,  $m = 8$ ,  $\Delta = 5$ , degrees  $[5, 3, 2, 2, 1, 1, 1, 1]$ ) every bound derivable from sequence data alone gives residue  $\leq 6$ : residue  $\leq \alpha = 6$ ; residue  $\leq n - \mu = 6$ ; and, from Lemma 2.1(2) with  $D_i \leq \Delta$ , residue  $\leq n - \lceil m/\Delta \rceil = 6$ . The truth is residue = 5, and (C61) needs exactly residue  $\leq f - 2 = 5$ . So the residual family demands a residue bound one better than every first-order degree-sequence estimate.

Two further routes were closed by proof rather than by witness. The ball-peeling certificate  $\Phi(G) := \max_{B'} (|B'| + \alpha(G - N[B']))$  over 2-packings  $B'$  satisfies  $\Phi(G) \leq \alpha(G)$  — for a 2-packing  $B'$  and an independent set  $I$  of  $G - N[B']$ , the set  $B' \cup I$  is independent in  $G$  — so it can never beat Fact 2.2 and dies exactly where O1 does; measured, it fails for 46.3% of the connected graphs with  $n \leq 8$ , the identical count to  $\alpha < \text{residue} + \lceil d/3 \rceil$ . And replacing  $\alpha$  by  $f$  in that recursion is vacuous, since  $\max_{B'} (|B'| + f(G - N[B'])) = f(G)$  attained at  $B' = \emptyset$ ; so the layering idea has to be grafted onto a maximum independent set, which is what Theorem 3.4 does.

## 4 The Havel–Hakimi toolkit

Everything in this section is proved under the *reductio* hypothesis alone, or with no hypothesis at all, and all of it is certified. The registry rows are R-1 to R-5 (and R-6, R-7 for the two corollaries mentioned in Remark 4.9); the evidence chain is a Qwen3.8-Max round, an independent Claude Opus 5 round, and a third, differently-briefed Opus session that confirmed the repairs of the second, with the campaign’s own reproduction of every joint (Appendix A).

Recall the vocabulary of Section 2: heads, survivors, blocks; and say an entry is *excess at step  $j$*  if its current value exceeds  $s - j + 1$ .

**Lemma 4.1** (survivor degree bound). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-12*] Under the *reductio hypothesis*, every survivor has degree  $\leq \tau$ ; equivalently, every vertex of degree  $\geq \tau + 1$  is a head, so  $\#\{v : \deg(v) \geq \tau + 1\} \leq \tau$ .

*Proof.* A survivor is decremented at most once per step and must reach 0 after  $\tau$  steps.  $\square$

**Lemma 4.2** (survivor decay, general  $s$ ). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-5*] Let  $s$  be the number of Havel–Hakimi steps. At the start of step  $j$ ,  $1 \leq j \leq s$ , every survivor has value  $\leq s - j + 1$ . (No *reductio hypothesis*.)

*Proof.* After step  $s$  every remaining entry is 0 by Lemma 2.1(1). A survivor is decremented at most once per step and only the  $s - j + 1$  steps  $j, j + 1, \dots, s$  remain.  $\square$

This lemma exists in the form stated because of a defect found in review. Its predecessor was stated for  $s = \tau$  only, yet was cited inside the proofs of Lemmas 4.4 and 4.5, which are asserted for general  $s$ . The statement of those two lemmas was true in the claimed generality all along; only the citation reached outside the cited lemma’s scope. Over 19 324 trajectories — the exhaustive connected atlas  $n \leq 7$  plus 4 500 random graphs  $n = 8, \dots, 12$ , canonical sort plus three adversarial tie-breaks — of which 14 784 have  $s \neq \tau$ , i.e. lie in exactly the regime where the old citation was invalid, Lemma 4.2 fails 0 times and Lemma 4.4 fails 0 times.

**Lemma 4.3** (terminal shape). [*certified: two review rounds, distinct model families, zero mathematics defects; rows R-11, R-12*]  $s = \tau$  if and only if the list at the start of step  $\tau$  is exactly  $[D_\tau, 1^{D_\tau}, 0^\infty]$ . Moreover, for an arbitrary graph (no *reductio*): if two entries are  $\geq 2$  at the start of step  $\tau$ , then  $s \neq \tau$ , whence by Fact 2.2  $s \geq \tau + 1$  and so  $\text{residue} \leq \alpha - 1$ .

*Proof.* The last step deletes the head and decrements the next  $D_\tau$  entries; all entries must then be 0, which is the displayed shape. For the second sentence, the displayed shape has exactly one entry  $\geq 2$  when  $D_\tau \geq 2$  and none when  $D_\tau = 1$ ; two entries  $\geq 2$  therefore contradicts it, so  $s \neq \tau$ , and  $s \geq \tau$  always by Fact 2.2.  $\square$

The second sentence is stated *outside* the *reductio* deliberately. Read inside the standing hypothesis  $s = \tau$  its antecedent contradicts the first sentence and it is vacuously true; the version above is the one every call site actually consumes. This distinction was itself the subject of a review finding.

**Lemma 4.4** (block occupancy). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-1*] At the start of step  $j$ , every later head whose current value exceeds  $s - j + 1$  lies in  $\text{block}_j$ , and is therefore decremented at step  $j$ .

*Proof.* Let  $x$  be deleted at step  $i > j$  with current value  $v > s - j + 1$ , and suppose  $x \notin \text{block}_j$ . The block is a prefix of the remaining entries in non-increasing order after deletion of the step- $j$  head, so  $v \leq \min(\text{block}_j)$ . If some block entry is a survivor then  $\min(\text{block}_j) \leq s - j + 1$  by Lemma 4.2, contradicting  $v > s - j + 1$ . Otherwise all  $D_j$  block entries are later heads; together with  $x$  that is  $D_j + 1$  later heads, while only  $s - j$  heads remain, so  $D_j \leq s - j - 1$ . But every block entry is at most the head value  $D_j$ , so  $v \leq \min(\text{block}_j) \leq D_j \leq s - j - 1 < s - j + 1$ , a contradiction.  $\square$

**Lemma 4.5** (head dichotomy). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-2*] Let a head be deleted at step  $i$ , of original degree  $g$ , and let  $h_i$  be the number of earlier steps at which it was decremented. Then:

- (a) persistence: if it is excess at step  $j < i$  then by Lemma 4.4 it is decremented at step  $j$ , hence it is excess at step  $j + 1$  as well;
- (b) if  $g \geq s + 1$  then it is excess at step 1, hence at every step  $1, \dots, i - 1$ , hence in every earlier block, so  $h_i = i - 1$  and  $D_i = g - (i - 1)$  exactly;
- (c) if  $g \leq s$  then  $D_i \leq s - i + 2$ , unconditionally.

*Proof.* (a) and (b) are immediate from Lemma 4.4 and the definition of excess. For (c): if the entry is never excess before deletion, its value at step  $i - 1$  is at most  $s - (i - 1) + 1 = s - i + 2$  and values never increase (for  $i = 1$ ,  $D_1 = g \leq s$ ). Otherwise let  $j_0 \geq 2$  be the first excess step; its value at step  $j_0 - 1$  is at most  $s - j_0 + 2$ , and by (a) it is decremented at each of the  $i - j_0$  steps  $j_0, \dots, i - 1$ , so  $D_i \leq (s - j_0 + 2) - (i - j_0) = s - i + 2$ .  $\square$

Lemma 4.5 is the engine of everything downstream. Clause (b) is an *equality*, and it is what makes the position sums of Section 7 collapse; clause (c) is an unconditional inequality for low heads. The pair replaces an earlier, hypothesis-laden head-decay lemma of the campaign, and is tie-safe, which the earlier one was not obviously.

**Theorem 4.6** (the  $K = B$  corner is a clique). [*certified*: two review rounds, distinct model families, zero mathematics defects; row R-3] Assume the reductio hypothesis  $\text{residue}(G) = \alpha(G)$  and that every  $b \in B$  has  $\deg(b) \geq \tau + 1$ . Then  $K = B$  and  $e_B = \binom{\tau}{2}$ :  $B$  is a clique.

*Proof.* By Lemma 4.1 and the hypothesis, every  $b \in B$  is a head; since  $|K| = s = \tau = |B|$  we get  $K = B$ , and hence the total number of decrements absorbed by heads before their deletion equals  $e_B$ . Every head now has original degree  $\geq \tau + 1 = s + 1$ , so Lemma 4.5(b) applies to each, giving  $h_i = i - 1$ . Summing,  $e_B = \sum_{i=1}^{\tau} (i - 1) = \binom{\tau}{2}$ .  $\square$

**Corollary 4.7.** [*certified*: two review rounds, distinct model families, zero mathematics defects; row R-4] Let  $G$  be connected with  $\text{diam}(G) = 4$ . If some maximum independent set  $A$  has  $\min_{b \in B} \deg(b) \geq \tau + 1$ , then  $\text{residue}(G) \leq \alpha(G) - 1$ .

*Proof.* Otherwise  $\text{residue} = \alpha$ , and Theorem 4.6 makes  $B$  a clique, contradicting Observation 5.2 below.  $\square$

*Remark 4.8* (independent numerical check). [computational: exhaustive  $n \leq 7$ , the exhaustive  $n = 8$  cover, and 6 000 denser random graphs  $n = 9, \dots, 12$ : 114 914 connected graphs, of which 1 464 have residue =  $\alpha$  with all  $B$ -degrees  $\geq \tau + 1$ ] In none of those 1 464 instances is  $B$  a non-clique. The check was written by the owner agent and reuses no code from the agent that proposed Theorem 4.6.

*Remark 4.9* (a statement of the campaign refuted by its own review). An earlier statement of this line asserted, unconditionally, that if every  $b \in B$  has  $\deg(b) \geq \tau + 1$  and  $e_B = 0$  then  $m \leq \Delta(G) + \tau(\tau + 1)/2 - 1$ . A reviewing session produced  $K_{2,3}$  as a counterexample: taking  $B$  to be the 2-element part,  $\alpha = 3$ ,  $\tau = 2$ ,  $e_B = 0$ , both  $B$ -degrees are  $3 = \tau + 1$ ,  $\Delta = 3$  and  $m = 6$ , while  $\Delta + \tau(\tau + 1)/2 - 1 = 5 < 6$ . The owner agent reproduced the witness and upheld the objection: the statement is true only under the reductio hypothesis — and indeed  $K_{2,3}$  has heads  $(3, 2, 1)$ , so  $s = 3 \neq \tau = 2$  and the reductio fails there, which is precisely why the hypothesis has to be written into the statement. The patched statement carries registry row R-7. Nothing downstream used the unconditional version. We report this because the honest measure of a review gate is what it catches, and because it is the smaller of the two refutations in this paper; the larger one is in Section 11.

## 5 Frame lemmas at diameter four

The four statements of this section hold in the hard-core frame (Definition 3.13) — no reduction hypothesis — and are the only structural input the later sections take from the geometry of  $G$ . All are certified as part of the root-reduction block of the closeout registry.

**Lemma 5.1** (pairing obstacle). [*certified: two review rounds, distinct model families, zero mathematics defects; root-reduction block*] Let  $A$  be a maximum independent set,  $B = V \setminus A$ , and suppose  $f = \alpha + 1$ . Then for distinct  $u, v \in B$ :

$$uv \in E \implies N_A(u) \cap N_A(v) \neq \emptyset, \quad uv \notin E \implies |N_A(u) \cap N_A(v)| \geq 2.$$

*Proof.* Consider  $G[A \cup \{u, v\}]$ , which has  $\alpha + 2$  vertices. If  $f = \alpha + 1$  it must contain a cycle. Since  $A$  is independent, every edge of that subgraph meets  $\{u, v\}$ , so a cycle is either the triangle  $u - v - a$  with  $a \in N_A(u) \cap N_A(v)$ , or the 4-cycle  $u - a - v - b$  with distinct  $a, b \in N_A(u) \cap N_A(v)$ . The first requires  $uv \in E$ .  $\square$

**Observation 5.2.** [*certified: two review rounds, distinct model families, zero mathematics defects; root-reduction block*] If  $\text{diam}(G) = 4$  then  $B$  is not a clique; equivalently  $\nu \geq 1$ .

*Proof.*  $A$  is a maximal independent set, so every  $a \in A$  has a neighbour in  $B$ . If  $B$  were a clique then for  $a, a' \in A$  we would have  $a - b - b' - a'$  for suitable  $b, b' \in B$ , so  $\text{dist}(a, a') \leq 3$ ; for  $a \in A, b \in B$  we would have  $\text{dist}(a, b) \leq 2$ ; and within  $B$ ,  $\text{dist} \leq 1$ . Hence  $\text{diam} \leq 3$ .  $\square$

**Fact 5.3** (the diameter witness). [*certified: two review rounds, distinct model families, zero mathematics defects; root-reduction block*] Call  $T \subseteq B$  an occurring type if  $T = N(a)$  for some  $a \in A$ . In the hard-core frame there exist occurring types  $T_1, T_2$  that are disjoint and have no  $G[B]$ -edge between them. In particular a singleton type occurs.

*Proof.* By Lemma 5.1, any two vertices of  $B$  have a common  $A$ -neighbour or are adjacent, so  $\text{dist}(b, b') \leq 2$  and hence  $\text{dist}(a, b) \leq 3$  for all  $a \in A, b \in B$ . So  $\text{diam} = 4$  can only be realised by a pair  $a, a' \in A$ , and  $\text{dist}(a, a') = 4$  holds if and only if  $N(a) \cap N(a') = \emptyset$  and there is no  $G[B]$ -edge between them.  $\square$

**Lemma 5.4** ( $\tau$ -uniform degree bound on the witness types). [*certified: two review rounds, distinct model families, zero mathematics defects; root-reduction block*] In the hard-core frame, let  $T_1, T_2$  realise Fact 5.3. Then every  $b \in T_1 \cup T_2$  has  $\deg_A(b) \geq 3$ , hence  $\deg(b) \geq 3 + \deg_B(b)$ .

*Proof.* Let  $b \in T_1$  and pick any  $b' \in T_2$ . There is no  $G[B]$ -edge between  $T_1$  and  $T_2$ , so  $bb' \notin E$ , and Lemma 5.1 gives  $|N_A(b) \cap N_A(b')| \geq 2$ . The witness  $a_1$  of type  $T_1$  is adjacent to  $b$  and, since  $b' \notin T_1$ , not adjacent to  $b'$ ; so  $a_1$  is none of those two common neighbours. Hence  $\deg_A(b) \geq 2 + 1 = 3$ . Symmetrically for  $b \in T_2$ .  $\square$

*Remark 5.5* (the measured obstacle). [computational: exhaustive  $n \leq 7$  plus the  $n = 8$  cover, restricted to the hard-core frame: 340 graphs, 399 pairs realising Fact 5.3, 0 failures of Lemma 5.4] The frame instances split by  $\tau$  as  $\{2 : 14, 3 : 200, 4 : 118\}$ , and the hypothesis of Corollary 4.7 — all  $B$ -degrees  $\geq \tau + 1$  — holds in 14 of 14, 91 of 200 and only 5 of 118 of them respectively. Lemma 5.4 explains the collapse: Fact 5.3 together with Lemma 5.1 forces  $\deg_A(b) \geq 3$  only for the  $O(1)$  vertices of  $T_1 \cup T_2$ , and  $\deg_A(b) \geq 2$  for any  $b$  with a non-neighbour in  $B$  — bounds that do not grow with  $\tau$ , while the hypothesis of Corollary 4.7 demands  $\tau + 1$ . This is why the  $K = B$  corner is a shrinking corner and why the rest of the paper is about  $K \neq B$ .

## 6 Small covers: $\tau \leq 3$

**Theorem 6.1** ( $\tau \leq 2$ ). [*certified: two review rounds, distinct model families, zero mathematics defects; closeout block “ $\tau \leq 2 / \tau = 3$ ”*] If  $\text{diam}(G) = 4$  and  $\tau(G) = n - \alpha(G) \leq 2$ , then (D4) holds for  $G$ .

*Proof.* We may assume  $f = \alpha + 1$ , since otherwise  $f \geq \alpha + 2$  and there is nothing to prove; so we are in the hard-core frame and Lemma 5.1 is available.

If  $\tau = 1$  then  $B$  is a single vertex covering all edges, so  $G$  is a star and  $\text{diam} \leq 2$ . Let  $\tau = 2$ ,  $B = \{x, y\}$ . Every  $a \in A$  has  $N(a) \subseteq \{x, y\}$  non-empty, so  $A$  partitions into  $A_x$  (type  $\{x\}$ ),  $A_y$  (type  $\{y\}$ ) and  $A_c$  (type  $\{x, y\}$ ); write  $a = |A_x|$ ,  $b = |A_y|$ ,  $c = |A_c|$ . By Fact 5.3 there are two disjoint occurring types with no  $G[B]$ -edge between them, which forces  $xy \notin E$  and  $A_x, A_y \neq \emptyset$ ; and then Lemma 5.1 applied to the non-adjacent pair  $x, y$  gives  $|A_c| = c \geq 2$ . Thus

$$\deg(x) = a + c, \quad \deg(y) = b + c, \quad m = a + b + 2c,$$

and the degrees of the  $A$ -vertices are 1 (on  $A_x \cup A_y$ ) or 2 (on  $A_c$ ).

Suppose residue  $= \alpha$ , i.e.  $s = \tau = 2$ , so  $D_1 + D_2 = m$  by Lemma 2.1(2). Assume without loss of generality  $a \geq b$ , so  $D_1 = \Delta = a + c$ . After the first step the entry  $y$  has value  $\deg(y) - 1$  or  $\deg(y)$  according to whether  $y \in \text{block}_1$ ; but  $\deg(y) = b + c \geq c \geq 2$  is the largest of the remaining entries, so  $y$  heads the sorted order and lies in  $\text{block}_1$  as soon as  $D_1 \geq 1$ , which holds. Hence  $D_2 \leq (b + c) - 1$  and

$$D_1 + D_2 \leq (a + c) + (b + c) - 1 = m - 1 < m,$$

a contradiction. Therefore  $s \geq 3$  and residue  $= n - s \leq n - 3 = \alpha - 1$ .  $\square$

**Theorem 6.2** ( $\tau = 3$ ). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-13, with repairs R1, R3, J1, J2, K1, K2, K3 landed*] If  $G$  is connected,  $\text{diam}(G) = 4$  and  $\tau(G) = 3$ , then (D4) holds for  $G$ . Combined with Theorem 6.1, (D4) holds whenever  $\tau \leq 3$ ; so the hard core has  $\tau \geq 4$ .

The proof is a finite case analysis on  $k := \#\{b \in B : \deg(b) \geq 4\}$ ; since every  $A$ -vertex has degree  $\leq \tau = 3$ , only  $B$ -vertices can have degree  $\geq 4$ . It is long, and we give its skeleton together with the tools it uses, rather than reproducing every branch; the complete branch-by-branch text, with the seven landed repairs listed in the marker, is in the campaign’s source report, and it is that text — not this summary — that the two review rounds certified. We flag this explicitly in Section 13 as the one place in the paper where a certified argument is summarised rather than reproduced.

*Proof skeleton.* Assume  $f = \alpha + 1$  and residue  $= \alpha$ , so  $s = 3$ ,  $\sum_{i \leq 3} D_i = m$ , and by Lemma 4.3 the list after step 2 is exactly  $[D_3, 1^{D_3}, 0^{\dots}]$ . Write  $B = \{x, y, z\}$  with  $B$ -degrees  $X \geq Y \geq Z$ , and for non-empty  $T \subseteq B$  let  $\text{mu}[T] = \#\{a \in A : N(a) = T\}$ ; put  $p = \text{mu}[B]$ ,  $q = \sum_{|T|=2} \text{mu}[T]$ ,  $r = \sum_{|T|=1} \text{mu}[T]$ , so  $\alpha = p + q + r$ ,  $n = \alpha + 3$ , and the  $A$ -degree spectrum is  $3^p 2^q 1^r$ . Since  $A$  is independent,  $G$  is determined up to isomorphism by these seven multiplicities together with the isomorphism type of  $G[B]$ , of which there are four; and  $e_B \leq 2$  by Observation 5.2. Three structural bounds follow from Lemma 5.1 and Fact 5.3:

- if  $e_B = 0$  then every  $b \in B$  has  $\deg(b) \geq 2$ ;
- if  $e_B = 1$ , say  $E(G[B]) = \{uv\}$  with third vertex  $w$ , then  $\deg(u), \deg(v), \deg(w) \geq 3$  and  $k \geq 1$ ;

- if  $e_B = 2$ , so  $G[B]$  is a path with centre  $c$  and ends  $e_1, e_2$ , then  $\deg(e_1), \deg(e_2) \geq 4$  — hence  $k \geq 2$  — and  $\deg(c) \geq 3$ .

Consequently  $k = 0 \Rightarrow e_B = 0$  and  $k = 1 \Rightarrow e_B \leq 1$ . One further tool is used: if the degree list with one copy of  $\Delta$  removed has all entries  $\leq 3$  and  $\Delta$  is at least the number of its entries equal to 3, then after step 1 all entries are  $\leq 2$ , so  $D_2, D_3 \leq 2$  and  $\sum_{i \leq 3} D_i \leq \Delta + 4$ ; hence  $m > \Delta + 4$  forces  $s \geq 4$ . Both hypotheses of this tool must be checked at each call site — the second is not automatic, and the campaign’s own control instance  $[3, 3, 3, 3, 3, 2, 1]$  (with  $\Delta = 3$  and  $m = 9 > 7$ ) is exactly where a blanket appeal to it would give a false bound. That over-general appeal was a defect found in review and removed; the tool is now invoked only at sites where its second hypothesis is verified in line.

The four branches then run as follows. For  $k = 3$  one gets  $D_1 = X$ ,  $D_2 = Y - 1$  and  $D_3 = Z + 1 - e_B$ , and comparing with the maximum possible value after step 2 forces  $e_B = 2$  and  $D_3 = Z - 1$  with  $Z = 4$ ; counting the entries of value  $\geq 2$  after step 2 then gives at least  $Y \geq 4$  of them, contradicting Lemma 4.3. For  $k = 2$  the three values of  $e_B$  are treated separately:  $e_B = 0$  forces  $Z \leq 2$  and an undecremented 3 at step 1, whence  $p \geq X \geq 4$  and so  $Z \geq p \geq 4$ , a contradiction;  $e_B = 1$  forces  $\deg_A(z) = p = 3$  and  $\text{mu}[\{x, z\}] = \text{mu}[\{y, z\}] = \text{mu}[\{z\}] = 0$ , which kills every pair admissible for Fact 5.3;  $e_B = 2$  pins the degree sequence to  $[a_0 + c_0 + 2, b_0 + c_0 + 2, 3, 3, 2^{c_0}, 1^{a_0+b_0}]$  with  $a_0, b_0, c_0 \geq 1$ , and the explicit trajectory gives  $\sum_{i \leq 3} D_i = m - 1 < m$ . For  $k = 1$  the two possible values of  $e_B$  each reduce to  $p \leq 2 \leq X - 2$ , where the tool above applies and contradicts the exact value of  $m$ ; in the sub-branch  $e_B = 0$ ,  $p = 3$  every occurring type contains one fixed vertex  $x$ , so every  $A$ -vertex is adjacent to  $x$ , every two  $A$ -vertices are at distance 2 through  $x$ ,  $\text{dist}(a, b) \leq 3$  and  $\text{dist}(b, b') = 2$ , whence  $\text{diam} \leq 3$  unconditionally — contradicting  $\text{diam} = 4$ . Finally  $k = 0$  forces  $e_B = 0$ , then  $p = 2$  and  $q = 1$ , and Fact 5.3 pins the graph up to isomorphism to the single instance  $A = \{u_1, u_2 \text{ universal}, w \sim \{y, z\}, \ell \sim \{x\}\}$ ,  $\deg = [3, 3, 3, 3, 3, 2, 1]$ ,  $n = 7$ ,  $m = 9$ ; running the process directly gives  $D = (3, 3, 2)$ ,  $\sum D_i = 8 = m - 1 < m$ , so  $s = 4$  and  $\text{residue} = 3 = \alpha - 1$ . These four cases are exhaustive, so  $\text{residue} = \alpha$  is impossible.  $\square$

*Remark 6.3* ( $\tau = 3$ : numerical backing, and non-vacuity). [computational: 591 710 hard-core-frame instances at  $\tau = 3$  inside a multiplicity box of size 5, and 147 348 instances inside a box of size 4 with the structural skeleton checked separately] The parameterisation above makes the  $\tau = 3$  frame an exhaustive scan rather than a search: every tuple in the box is covered, no isomorphism rejection is needed, and no SAT solver is involved. In the size-5 box there are 0 instances with  $\text{residue} \geq \alpha$ ; the distribution by  $e_B$  is  $\{0 : 224\,035, 1 : 196\,075, 2 : 171\,600\}$  with  $e_B = 3$  never occurring, which is a numerical confirmation of Observation 5.2; and the minimum of  $m - (D_1 + D_2 + D_3)$  is exactly 1, attained at  $B$  independent,  $A = \{u_1, u_2 \text{ universal}, w \sim \{x, y\}, e \sim \{z\}\}$ ,  $n = 7$ ,  $m = 9$ ,  $\deg = [3, 3, 3, 3, 3, 2, 1]$ . In the size-4 box the  $(e_B, k)$  cells  $(1, 0)$ ,  $(2, 0)$  and  $(2, 1)$  are always empty, confirming the two implications  $e_B = 1 \Rightarrow k \geq 1$  and  $e_B = 2 \Rightarrow k \geq 2$  used above. Independently, the round-B judge enumerated every  $\tau = 3$  graph on  $n \leq 37$  (89.8M parameter tuples), a further 77.9M-tuple box up to  $n = 73$ , 7.7M huge-multiplicity instances with  $n$  up to about  $10^6$ , and a  $2^{21}$  brute force at  $n = 7$  reproducing exactly the five hard-core degree sequences its own parameterisation predicts, with no counterexample anywhere — and, decisively for the question of vacuity, 24 013 near-misses: configurations satisfying every hard-core condition except  $f = \alpha + 1$ . Theorem 6.2 is therefore excluding a class that is approached from every direction but one, not quantifying over an empty set.

*Remark 6.4* (a negative result worth recording). [computational: the tight instance of Remark 6.3] The natural entry-wise bound  $D_i \leq d_i - (i - 1)$  — which, if true, would close the whole problem by a one-line summation — already fails *inside* the hard core: for  $\deg = (3, 3, 3, 3, 3, 2, 1)$  one has

$D = (3, 3, 2)$ , so  $d_2 - D_2 = 0$ . Moreover the sum that a proof of this shape would need is tight in that same instance. Any argument resting only on entry-wise decrement estimates is therefore dead, which is why Lemma 4.5 proceeds instead by bounding head values through survivor decay.

## 7 A $\tau$ -uniform budget

From here on the parameter that organises everything is  $L = |B_{\text{lo}}|$ , the number of vertices of  $B$  of degree at most  $\tau$ . Theorem 4.6 says the hard core has  $L \neq 0$ ; this section proves  $L \geq 2$  and Section 8 raises it to  $L \geq 3$ . The results of this section carry registry rows R-16, R-17 and R-18. Note that the high/low split is by *degree*, not by head status; Lemma 7.1 makes the two agree on  $B_{\text{hi}}$  only, and it is exactly that asymmetry which lets Theorem 7.2 say something when  $K \neq B$ .

### 7.1 The bound

**Lemma 7.1.** [*certified: two review rounds, distinct model families, zero mathematics defects; row R-16, tier “reductio only”*] Under the reductio hypothesis,  $B_{\text{hi}}$  is exactly the set of heads of original degree  $\geq \tau + 1$ ; in particular there are  $p$  such heads and  $L$  heads of original degree  $\leq \tau$ . Write  $I_{\text{hi}}$ ,  $I_{\text{lo}}$  for the corresponding sets of head positions, so  $|I_{\text{hi}}| = p$  and  $|I_{\text{lo}}| = L$ .

*Proof.* Every  $a \in A$  has  $N(a) \subseteq B$ , so  $\deg(a) \leq |B| = \tau$ ; hence every vertex of degree  $\geq \tau + 1$  lies in  $B$ , and by Lemma 4.1 every such vertex is a head.  $\square$

**Theorem 7.2** (the budget inequality). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-16, tier “reductio only”*] Under the reductio hypothesis  $\text{residue}(G) = \alpha(G)$ , with no hypothesis on  $K$ :

$$\sum_{b \in B_{\text{lo}}} \deg(b) + \nu \leq L(\tau + 1); \quad (7.1)$$

$$\sum_{b \in B_{\text{lo}}} \deg_A(b) + e(B_{\text{lo}}) + \bar{m} \leq \frac{1}{2}L(L + 3); \quad (7.2)$$

$$\sum_{b \in B_{\text{lo}}} \deg_A(b) + \bar{m} \leq 2L + \nu(B_{\text{lo}}). \quad (7.3)$$

*Proof.* By Lemma 2.1(2),  $\sum_{i=1}^{\tau} D_i = m$ . Split the head positions by Lemma 7.1. For  $i \in I_{\text{hi}}$ , Lemma 4.5(b) gives  $D_i = g_i - (i - 1)$  exactly, where  $g_i$  is the original degree; for  $i \in I_{\text{lo}}$ , Lemma 4.5(c) gives  $D_i \leq \tau - i + 2$ . Hence

$$m \leq \sum_{b \in B_{\text{hi}}} \deg(b) - \sum_{i \in I_{\text{hi}}} (i - 1) + \sum_{i \in I_{\text{lo}}} (\tau - i + 2).$$

On the other hand  $m = \sum_{b \in B} \deg(b) - e_B$ . Subtracting,

$$\sum_{b \in B_{\text{lo}}} \deg(b) - e_B \leq - \sum_{i \in I_{\text{hi}}} (i - 1) + \sum_{i \in I_{\text{lo}}} (\tau - i + 2).$$

The right-hand side is independent of which positions the low heads occupy: adding and subtracting  $\sum_{i \in I_{\text{lo}}} (i - 1)$ ,

$$- \sum_{i \in I_{\text{hi}}} (i - 1) + \sum_{i \in I_{\text{lo}}} (\tau - i + 2) = - \sum_{i=1}^{\tau} (i - 1) + \sum_{i \in I_{\text{lo}}} [(i - 1) + (\tau - i + 2)] = - \binom{\tau}{2} + L(\tau + 1),$$

because the bracket collapses to the constant  $\tau + 1$ . With  $e_B = \binom{\tau}{2} - \nu$  this is (7.1). For (7.2) write  $\sum_{B_{lo}} \deg(b) = \sum_{B_{lo}} \deg_A(b) + 2e(B_{lo}) + e(B_{lo}, B_{hi})$  and  $\nu = \nu(B_{lo}) + \nu(B_{lo}, B_{hi}) + \bar{m}$  with  $\nu(B_{lo}) = \binom{L}{2} - e(B_{lo})$  and  $\nu(B_{lo}, B_{hi}) = Lp - e(B_{lo}, B_{hi})$ ; the crossing terms cancel and  $L(\tau + 1) - \binom{L}{2} - Lp = \frac{1}{2}L(L + 3)$ . Finally (7.3) is (7.2) with  $e(B_{lo}) = \binom{L}{2} - \nu(B_{lo})$ .  $\square$

Theorem 4.6 is the case  $L = 0$ : then (7.1) reads  $\nu \leq 0$ , i.e.  $B$  is a clique. So Theorem 7.2 subsumes it, and it never mentions  $K$  at all.

**Theorem 7.3** (slack positivity). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-16, tier “reductio only”*] Assume the reductio hypothesis and  $\tau \geq 2$ . If  $L \geq 1$  then

$$\text{slack} := L(\tau + 1) - \left( \sum_{b \in B_{lo}} \deg(b) + \nu \right) \geq 1,$$

i.e. (7.1) holds with one unit to spare; equivalently  $\sum_{b \in B_{lo}} \deg_A(b) + e(B_{lo}) + \bar{m} \leq \frac{1}{2}L(L + 3) - 1$ .

*Proof.* From the proof of Theorem 7.2,  $\text{slack} = \sum_{i \in I_{lo}} (\tau - i + 2 - D_i)$ , each summand being  $\geq 0$  by Lemma 4.5(c). Suppose  $\text{slack} = 0$ ; then

$$\text{every low head satisfies } D_i = \tau - i + 2 \text{ exactly.} \quad (*)$$

Let  $i_0 := \max I_{lo}$  and let  $x$  be the head deleted at step  $i_0$ , of original degree  $g = \deg(x) \leq \tau$ . Write  $v_k$  for  $x$ 's value at the start of step  $k$ .

*Step 1:*  $i_0 \geq 2$ . If  $i_0 = 1$  then  $I_{lo} = \{1\}$ , so every head at a position  $2, \dots, \tau$  is high, of original degree  $\geq \tau + 1$ ; but  $D_1 = \Delta = g \leq \tau$  by Lemma 2.1(2), contradicting that  $\Delta$  is the maximum degree. (If  $L = \tau$  there are no high heads and  $i_0 = \tau \geq 2$  directly.)

*Step 2:* an escape step exists, and it sits exactly at the decay threshold. Let  $\mathcal{E} := \{k < i_0 : x \notin \text{block}_k\}$ . Since  $v_{i_0} = g - (i_0 - 1 - |\mathcal{E}|)$  and  $v_{i_0} = D_{i_0} = \tau - i_0 + 2$  by  $(*)$ , we get  $|\mathcal{E}| = \tau + 1 - g \geq 1$ . Put  $j := \max \mathcal{E}$ . From step  $j + 1$  to step  $i_0 - 1$  the entry  $x$  is decremented every time, so  $v_{i_0} = v_j - (i_0 - 1 - j)$ , whence  $v_j = \tau - j + 1$ .

*Step 3:* what  $\text{block}_j$  can contain.  $\text{block}_j$  is the prefix of the  $D_j$  largest non-head entries in non-increasing order, and  $x$  is an entry of step  $j$  lying outside it; hence every entry of  $\text{block}_j$  has value  $\geq v_j = \tau - j + 1$ . By Lemma 4.2 every survivor has value  $\leq \tau - j + 1$  at the start of step  $j$ . Therefore each survivor in  $\text{block}_j$  has value exactly  $\tau - j + 1$  and so — needing  $\tau - j + 1$  further decrements in the  $\tau - j + 1$  remaining steps — lies in  $\text{block}_k$  for every  $k \in [j, \tau]$ . Write  $s_j$  for the number of survivors in  $\text{block}_j$ .

*Step 4:* the head part of  $\text{block}_j$ . Every entry of  $\text{block}_j$  is either a survivor or a head deleted later. Let  $\ell_j := \#\{i \in I_{lo} : i > j\}$  and note  $x$  is one of these and  $x \notin \text{block}_j$ , so the low later heads contribute  $\lambda_j \leq \ell_j - 1$ . Every high later head lies in every earlier block by Lemma 4.5(b), and there are  $(\tau - j) - \ell_j$  of them. Hence  $D_j = |\text{block}_j| = (\tau - j - \ell_j) + \lambda_j + s_j$ .

*Step 5:*  $s_j \leq 2$ . All heads after position  $i_0$  are high, so by Lemma 4.5(b) all  $\tau - i_0$  of them lie in  $\text{block}_{i_0}$ ; by Step 3 the  $s_j$  survivors lie in  $\text{block}_{i_0}$  too, and these sets are disjoint. Hence  $D_{i_0} \geq s_j + (\tau - i_0)$ , and  $D_{i_0} = \tau - i_0 + 2$  gives  $s_j \leq 2$ .

*Step 6:* contradiction. Steps 4 and 5 give  $D_j \leq (\tau - j - \ell_j) + (\ell_j - 1) + 2 = \tau - j + 1$ . But position  $j < i_0$  is a head position, and either it is high, so  $D_j = g_j - (j - 1) \geq (\tau + 1) - (j - 1) = \tau - j + 2$  by Lemma 4.5(b), or it is low, so  $D_j = \tau - j + 2$  by  $(*)$ . Either way  $D_j \geq \tau - j + 2 > \tau - j + 1 \geq D_j$ .  $\square$

*Remark 7.4* (sharpness). [computational: 2740 trajectories on the exhaustive atlas  $n \leq 7$  and 19710 on 9000 random graphs  $n = 8, \dots, 13$  plus 4000 near-split graphs, five trajectories each] Theorem 7.3 is *not*

the statement “the last low head satisfies  $D_{i_0} \leq \tau - i_0 + 1$ ”: that stronger claim is false, with 1 925 counterexamples in the corpus. What Step 6 extracts is that the escape step  $j$  must itself be a low head position carrying strict inequality, so equality cannot hold at all low heads simultaneously — which is slack  $\geq 1$  and no more. Every intermediate assertion of the proof was checked separately, not just the conclusion, with 0 failures, and the histogram of  $s_j$  at equality heads is  $\{2 : 2300\}$ : Step 5’s bound is attained every time it is used. In a separate corpus of 6 403 graphs and 34 040 reduction trajectory checks, the minimum slack of (7.1) per  $L$  is  $\{0 : 0, 1 : 1, 2 : 1, 3 : 2, 4 : 2, 5 : 3, \dots\}$ : tight exactly at  $L = 0$ , which is forced by Theorem 4.6, and never 0 for  $L \geq 1$ .

**Corollary 7.5** (the hard core has  $L \geq 2$ ). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-18] In the hard core, at least two vertices of  $B$  have degree  $\leq \tau$ .*

*Proof.*  $L = 0$  is impossible: Theorem 4.6 would make  $B$  a clique, contradicting Observation 5.2.  $L = 1$  is impossible by Corollary 7.7 below (or, by the longer route of Proposition 7.9 and Corollary 7.10, because the slack would be exactly 0, contradicting Theorem 7.3). The numerical hypothesis  $\tau \geq 2$  of Theorem 7.3 is supplied by  $\text{diam} = 4$  alone: a connected graph with  $\tau \leq 1$  has a vertex cover of size  $\leq 1$ , hence is edgeless or a star, hence has  $\text{diam} \leq 2$ .  $\square$

The last sentence of that proof is worth a word. In three successive states of this text the parenthetical there cited  $\tau \geq 3$ , then  $\tau \geq 4$ , then Theorem 6.2 — each time a true statement that the proof does not consume, and each time making a self-contained corollary depend on the heaviest theorem in the line, hence unreviewable by any judge not handed that theorem. A review round objected, and the elementary local replacement above is the repair. The species — *a true import nobody needs* — recurred six times in the campaign’s source report and is the single most common defect its review gate caught.

## 7.2 The master budget

**Theorem 7.6** (master budget). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-17] In the hard core, for  $L \geq 1$ ,*

$$|B_{\text{lo}}^+| + c + \bar{m} \leq L + \nu(B_{\text{lo}}) - 1, \quad \text{where } c := |B_{\text{lo}} \cap (T_1 \cup T_2)|.$$

*Proof.* Lemma 5.1 gives  $\deg_A(b) \geq 2$  for every  $b \in B_{\text{lo}}^+$ , and  $\deg_A(b) \geq 1$  for every  $b \in B_{\text{lo}}$  since  $A$  is a maximum independent set. Lemma 5.4 gives  $\deg_A(b) \geq 3$  for  $b \in T_1 \cup T_2$ , and every such  $b$  that is low lies in  $B_{\text{lo}}^+$  — it has the whole of the other type among its non-neighbours — so  $B_{\text{lo}} \cap (T_1 \cup T_2) \subseteq B_{\text{lo}}^+$  and  $c \leq |B_{\text{lo}}^+|$ . Hence

$$\sum_{b \in B_{\text{lo}}} \deg_A(b) \geq (L - |B_{\text{lo}}^+|) \cdot 1 + (|B_{\text{lo}}^+| - c) \cdot 2 + c \cdot 3 = L + |B_{\text{lo}}^+| + c.$$

Substituting into the (7.3) form of Theorem 7.3,  $\sum_{b \in B_{\text{lo}}} \deg_A(b) + \bar{m} \leq 2L + \nu(B_{\text{lo}}) - 1$ , gives the claim.  $\square$

The restriction “that is low” in the third sentence is not decoration: the corresponding sentence without it is false for a *high*  $b \in T_1 \cup T_2$ , since  $B_{\text{lo}}^+ \subseteq B_{\text{lo}}$ , and Proposition 7.11(d) below is its own witness. That was a review finding; the count is unaffected, since it quantifies only over  $b \in B_{\text{lo}}$ .

**Corollary 7.7.** *[certified: two review rounds, distinct model families, zero mathematics defects; row R-17] The hard core has no instance with  $L = 1$ .*

*Proof.*  $L = 1$  gives  $\nu(B_{\text{lo}}) = 0$ , so Theorem 7.6 reads  $|B_{\text{lo}}^+| + c + \bar{m} \leq 0$ : the unique low vertex  $b_0$  is  $B$ -universal,  $c = 0$  and  $\bar{m} = 0$ .  $B$ -universality of  $b_0$  means no non-edge of  $B$  meets  $b_0$ , so  $\nu = \bar{m} = 0$  and  $B$  is a clique, contradicting Observation 5.2.  $\square$

**Corollary 7.8** (a floor on  $L$ ). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-17*] *In the hard core, if every low vertex is  $B$ -universal then  $\nu = \bar{m} \leq L - 1$ , so  $L \geq \nu + 1 \geq 2$ .*

*Proof.*  $B$ -universality of all of  $B_{\text{lo}}$  gives  $\nu(B_{\text{lo}}) = \nu(B_{\text{lo}}, B_{\text{hi}}) = 0$ , hence  $\nu = \bar{m}$ , and  $c = 0$  since a  $B$ -universal vertex has no non-neighbour and so cannot lie in  $T_1 \cup T_2$ . Theorem 7.6 then reads  $\bar{m} \leq L - 1$ , and  $\nu \geq 1$  by Observation 5.2.  $\square$

### 7.3 The two rigid low configurations

The following two statements were the campaign's original route to  $L \geq 2$ ; they are superseded in part by Corollary 7.7 and Theorem 9.3 but are certified in their own right, and Proposition 7.11 is what identifies the configuration that Section 8 then eliminates.

**Proposition 7.9** ( $L = 1$  is rigid). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-17*] *In the hard core with  $L = 1$ , write  $B_{\text{lo}} = \{b_0\}$ . Then  $b_0$  is adjacent to every other vertex of  $B$ ,  $\deg_A(b_0) = 1$ ,  $\deg(b_0) = \tau$  and  $\nu = \bar{m} = 1$ ; the unique non-adjacent pair of  $B$  is a pair  $u, v \in B_{\text{hi}}$  with  $T_1 = \{u\}$ ,  $T_2 = \{v\}$  and  $\deg_A(u), \deg_A(v) \geq 3$ ; and the unique  $A$ -neighbour  $a_0$  of  $b_0$  is adjacent to all of  $B$ .*

*Proof.*  $L = 1$  gives  $\nu(B_{\text{lo}}) = 0$ , so (7.3) of Theorem 7.3 reads  $\deg_A(b_0) + \bar{m} \leq 2$ . If  $b_0 \in T_1 \cup T_2$  then Lemma 5.4 gives  $\deg_A(b_0) \geq 3$ , impossible; so  $T_1, T_2 \subseteq B_{\text{hi}}$ . If  $b_0$  had a non-neighbour in  $B$ , Lemma 5.1 would give  $\deg_A(b_0) \geq 2$ , hence  $\bar{m} = 0$ , i.e.  $B_{\text{hi}}$  is a clique; but  $T_1, T_2$  are disjoint non-empty subsets of  $B_{\text{hi}}$ , so some  $u \in T_1$  and  $v \in T_2$  are adjacent, contradicting Fact 5.3. So  $b_0$  is adjacent to all of  $B \setminus \{b_0\}$ , whence  $\nu = \bar{m}$ ; by Observation 5.2  $\nu \geq 1$ , so  $\deg_A(b_0) + \bar{m} \leq 2$  forces  $\deg_A(b_0) = 1$  and  $\nu = \bar{m} = 1$ , and then  $\deg(b_0) = (\tau - 1) + 1 = \tau$ . Since the only non-adjacent pair of  $B$  is that single pair  $uv \subseteq B_{\text{hi}}$ , and all cross pairs of  $T_1 \times T_2$  must be non-adjacent,  $|T_1| = |T_2| = 1$  and  $\{T_1, T_2\} = \{\{u\}, \{v\}\}$ ; Lemma 5.4 gives  $\deg_A(u), \deg_A(v) \geq 3$ . Finally, for each  $w \in B \setminus \{b_0\}$  the pair  $(b_0, w)$  is adjacent, so Lemma 5.1 gives a common  $A$ -neighbour, which must be  $a_0$ .  $\square$

**Corollary 7.10.** [*certified: two review rounds, distinct model families, zero mathematics defects; row R-17*] *In the hard core with  $L = 1$  the slack of (7.1) is exactly 0:  $\sum_{b \in B_{\text{lo}}} \deg(b) + \nu = \tau + 1 = L(\tau + 1)$ .*

*Proof.* Immediate from Proposition 7.9:  $\deg(b_0) = \tau$  and  $\nu = 1$ .  $\square$

**Proposition 7.11** ( $L = 2$  is rigid). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-17*] *Suppose the hard core has an instance with  $L = 2$ ,  $B_{\text{lo}} = \{b_1, b_2\}$ . Then:*

- (a)  $b_1 \sim b_2$ ;
- (b) both are  $B$ -universal;
- (c)  $\deg_A(b_1) = \deg_A(b_2) = 1$  and their unique  $A$ -neighbours coincide in a single vertex  $a_0$  adjacent to all of  $B$ ;
- (d)  $\nu = \bar{m} = 1$ , i.e.  $G[B] = K_\tau$  minus exactly one edge  $uv$  with  $u, v \in B_{\text{hi}}$ , and  $T_1 = \{u\}$ ,  $T_2 = \{v\}$ ,  $\deg_A(u), \deg_A(v) \geq 3$ ;
- (e) the slack of (7.1) is exactly 1, so Theorem 7.3 is tight here.

*Proof.* Theorem 7.6 reads  $|B_{\text{lo}}^+| + c + \bar{m} \leq 1 + \nu(B_{\text{lo}})$ .

(a) If  $b_1 \not\sim b_2$  then  $\nu(B_{\text{lo}}) = 1$  and both lie in  $B_{\text{lo}}^+$ , so  $2 + c + \bar{m} \leq 2$ , giving  $c = \bar{m} = 0$ . Then  $B_{\text{hi}}$  is a clique and  $T_1, T_2 \subseteq B_{\text{hi}}$  are disjoint and non-empty, so some  $u \in T_1$  is adjacent to some  $v \in T_2$ , contradicting Fact 5.3. Hence  $b_1 \sim b_2$  and  $\nu(B_{\text{lo}}) = 0$ , so  $|B_{\text{lo}}^+| + c + \bar{m} \leq 1$ .

(b) If some  $b_i \in B_{\text{lo}}^+$  then  $c = \bar{m} = 0$  and the same contradiction recurs. Hence  $B_{\text{lo}}^+ = \emptyset$  and  $c = 0$ , leaving  $\bar{m} \leq 1$ .

(d) Both  $b_i$  being  $B$ -universal, every non-edge of  $B$  lies inside  $B_{\text{hi}}$ , so  $\nu = \bar{m} \leq 1$ ; Observation 5.2 gives  $\nu \geq 1$ , so  $\nu = \bar{m} = 1$ . The unique non-adjacent pair of  $B$  is  $uv \subseteq B_{\text{hi}}$ ; since all cross pairs of  $T_1 \times T_2$  are non-adjacent,  $T_1 = \{u\}$  and  $T_2 = \{v\}$ , and Lemma 5.4 gives  $\deg_A(u), \deg_A(v) \geq 3$ .

(c) At  $L = 2$ ,  $|B_{\text{lo}}^+| = c = 0$ ,  $\bar{m} = 1$ ,  $\nu(B_{\text{lo}}) = 0$ , both bounds combined in the proof of Theorem 7.6 are simultaneously tight: the degree count gives the lower bound  $\sum_{b \in B_{\text{lo}}} \deg_A(b) \geq L + |B_{\text{lo}}^+| + c = 2$ , while (7.3) of Theorem 7.3 gives the upper bound  $\sum_{b \in B_{\text{lo}}} \deg_A(b) \leq 2L + \nu(B_{\text{lo}}) - 1 - \bar{m} = 2$ . The two meet, so  $\deg_A(b_1) = \deg_A(b_2) = 1$ . Since  $b_1 \sim b_2$ , Lemma 5.1 supplies a common  $A$ -neighbour, necessarily the unique  $A$ -neighbour of each; call it  $a_0$ . For every  $w \in B \setminus \{b_1\}$  the pair  $(b_1, w)$  is adjacent, so Lemma 5.1 gives a common  $A$ -neighbour, necessarily  $a_0$ ; hence  $a_0$  is adjacent to all of  $B$ .

(e)  $\deg(b_i) = (\tau - 1) + 1 = \tau$ , so  $\text{slack} = 2(\tau + 1) - (2\tau + \nu) = 1$ .  $\square$

Writing the two bounds of (c) out as a squeeze, rather than saying only “re-run the degree count”, was a review repair: a tightness argument needs the inequality chain to be tight at every link, and naming one of the two links is not enough.

## 8 The fan family

Proposition 7.11 says that the  $L = 2$  layer of the hard core, if non-empty, is a single rigid configuration, on which Theorem 7.3 is tight and therefore cannot help. This section eliminates that configuration — and its whole family — by a Havel–Hakimi argument.

**Definition 8.1** (the fan configurations). For  $\tau \geq 2$ ,  $L \geq 1$  and  $\nu \geq 1$ , say that  $G$  carries the configuration  $\text{GFan}(\tau, L, \nu)$  if:  $|B| = \tau$ ; the  $L$  vertices of  $B_{\text{lo}}$  are all  $B$ -universal, each with  $\deg_A = 1$ , and their common  $A$ -neighbour is a single vertex  $a_0$  adjacent to all of  $B$ ; all  $\nu$  non-edges of  $B$  lie inside  $B_{\text{hi}}$ ; and every  $a \in A' := A \setminus \{a_0\}$  has  $N(a) \subseteq B_{\text{hi}}$ . Write  $\text{Fan}(\tau, L) := \text{GFan}(\tau, L, 1)$ , so that  $\text{Fan}(\tau, L)$  is the configuration of Proposition 7.11 at general  $L$ :  $G[B] = K_\tau$  minus one edge  $uv$  with  $u, v \in B_{\text{hi}}$ . Throughout put  $C := B_{\text{lo}} \cup \{a_0\}$  and  $p := \tau - L = |B_{\text{hi}}|$ .

In  $\text{Fan}(\tau, L)$  every  $c \in C$  has  $\deg(c) = \tau$  exactly ( $\tau - 1$  inside  $B$  plus  $a_0$  for  $c \in B_{\text{lo}}$ ; and  $a_0$  has exactly its  $\tau$  neighbours in  $B$ ); every  $w \in B_{\text{hi}}$  has  $\deg(w) \geq \tau + 1$ ; and every  $a \in A'$  has  $\deg(a) \leq p$ . Note  $u, v \in B_{\text{hi}}$  forces  $p \geq 2$ , i.e.  $L \leq \tau - 2$ .

We call steps  $1, \dots, p$  the *high phase*, and say a vertex *escapes*  $\text{block}_t$  if it is not decremented at step  $t$ ;  $E$  denotes the total number of (vertex, step) escapes of  $C$ -vertices during the high phase.

**Lemma 8.2** (high phase first). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-9*] In  $\text{Fan}(\tau, L)$  under the *reductio hypothesis*, the first  $p$  heads may be taken to be exactly the  $p$  vertices of  $B_{\text{hi}}$ .

*Proof.* Swapping the roles of two entries of equal current value — one taken as head, the other as a block member — leaves the value multiset trajectory unchanged, hence leaves  $s$  unchanged; so the *reductio hypothesis* is tie-break invariant and we may fix the tie-break. Induct on  $j \leq p$ : if

all heads before  $j$  are high, then  $p - (j - 1) \geq 1$  high vertices remain, each in every earlier block by Lemma 4.5(b), hence of value  $\deg - (j - 1) \geq \tau - j + 2$ , so  $D_j \geq \tau - j + 2$ . A low head at  $j$  has  $D_j \leq \tau - j + 2$  by Lemma 4.5(c); equality is a tie with a remaining high vertex, and the fixed tie-break takes the high vertex.  $\square$

**Lemma 8.3** (the endgame is unconditional). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-9] If at the start of some step the remaining multiset is  $[L]^{L+1}, 1, 1$  with  $L \geq 2$ , then the process needs exactly  $L + 1$  further steps.*

*Proof.* Head values are non-increasing. At the first of these steps the maximum is  $L$  and the  $L$  other  $L$ -entries are the top  $L$  non-head entries, so the block is exactly those, they become  $L - 1$ , and the two 1s survive untouched: the list becomes  $[L - 1]^L, 1, 1$ . Inductively after  $j$  steps the list is  $[L - j]^{L+1-j}, 1, 1$ , so after  $L - 1$  steps it is  $[1]^2, 1, 1 = [1, 1, 1, 1]$ : the head is 1,  $D = 1$ , one entry is zeroed and  $[0, 1, 1]$  remains. One further step clears it. That is  $L + 1$  steps in total.  $\square$

**Lemma 8.4** (the residue mass). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-9] In  $\text{Fan}(\tau, L)$  under the reductio hypothesis, assume  $E = 0$ . Then the  $A'$  entries at the start of step  $p + 1$  sum to exactly 2.*

*Proof.* By Lemma 8.2 the heads of steps  $1, \dots, p$  are  $B_{\text{hi}}$ , so by Lemma 4.5(b),  $\sum_{j \leq p} D_j = \sum_{x \in B_{\text{hi}}} \deg(x) - \binom{p}{2}$ . Now  $\sum_{x \in B_{\text{hi}}} \deg_B(x) = 2(\tau - 2) + (p - 2)(\tau - 1) = p\tau - p - 2$ , since  $u, v$  miss the edge  $uv$  and the other  $p - 2$  vertices of  $B_{\text{hi}}$  do not, and  $\sum_{x \in B_{\text{hi}}} \deg_A(x) = p + R$  where  $R := \sum_{a \in A'} \deg(a)$  and the  $p$  counts  $a_0$ 's edges to  $B_{\text{hi}}$ ; so  $\sum_{B_{\text{hi}}} \deg = p\tau - 2 + R$ . Counting the same decrements by recipient: a high head deleted at position  $i \leq p$  receives  $i - 1$  of them, total  $\binom{p}{2}$ ; each of the  $L + 1$  vertices of  $C$  receives  $p$  by hypothesis, total  $p(L + 1)$ ; the rest go to  $A'$ . Hence  $\text{dec}_{A'} = (p\tau - 2 + R - \binom{p}{2}) - \binom{p}{2} - p(L + 1) = R - 2$ , and  $A'$  starts at total  $R$ , leaving total 2.  $\square$

So the residual  $A'$  partition is either  $1 + 1$  or a single 2, and nothing else is arithmetically possible. Which of the two occurs decides the whole family:

**Observation 8.5.** *[certified: two review rounds, distinct model families, zero mathematics defects; row R-14] For every  $L \geq 2$ , the multiset  $[L]^{L+1}, 1, 1$  clears in  $L + 1$  Havel–Hakimi steps, while  $[L]^{L+1}, 2$  clears in exactly  $L$ .*

*Proof.* The first half is Lemma 8.3. The second is Lemma 8.12 below at  $\lambda = [2]$ : there  $\lambda_1 = 2$  and  $s_0(\lambda) = \text{steps}([2]^3 \cup [2]) = 2 = \lambda_1$ , so the total is  $(L - 2) + 2 = L$  for every  $L \geq 2$ .  $\square$

Since the reductio hypothesis requires exactly  $L$  further steps after step  $p$ , an instance whose  $A'$  residue were a single 2 would be a genuine survivor. That is the single load-bearing point of the whole elimination, at every  $L \geq 2$  — not a technicality of  $L = 2$ , as an earlier reading had it.

**Lemma 8.6** (escape budget). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-9] In  $\text{Fan}(\tau, L)$  under the reductio hypothesis,  $E \leq 1$ .*

*Proof.* Repeating the count of Lemma 8.4 with  $p(L + 1) - E$  decrements delivered to  $C$  gives  $\text{dec}_{A'} = R - 2 + E$ , so the  $A'$  residue is  $2 - E \geq 0$ , whence  $E \leq 2$ . If  $E = 2$  the  $A'$  residue is 0 and the  $C$ -part at the start of step  $p + 1$  is  $[L + 2, L^L]$  or  $[L + 1, L + 1, L^{L-1}]$ ; in both, the head exceeds the number of positive remaining entries ( $L + 2 > L$ , resp.  $L + 1 > L$ ), so the next step would drive a zero entry negative, which is impossible for the degree sequence of a graph.  $\square$

**Lemma 8.7** (no escape at all). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-9] In  $\text{Fan}(\tau, L)$  with  $L \geq 2$ , under the reductio hypothesis,  $E = 0$ .*

*Proof.* Suppose  $E \geq 1$  and let  $t \leq p$  be a step at which  $e_t \geq 1$  vertices of  $C$  escape; let  $c$  be one of them. By Lemma 8.2 the head of step  $t$  is a  $B_{\text{hi}}$  vertex, so  $D_t = g_t - (t - 1) \geq \tau - t + 2$ ; by Lemma 4.5(b) the  $p - t$  remaining  $B_{\text{hi}}$  vertices lie in  $\text{block}_t$ , and so does each of the  $(L + 1) - e_t$  non-escaping  $C$ -vertices. Hence  $|\text{block}_t \cap A'| = D_t - (p - t) - (L + 1) + e_t \geq (\tau - t + 2) - (p - t) - (L + 1) + 1 = 2$ , so  $\text{block}_t$  contains some  $x \in A'$ . Because  $\text{block}_t$  is a prefix of the sorted list and  $c \notin \text{block}_t$ , we get  $v_t(x) \geq v_t(c) \geq \tau - t + 1$ . By the count in Lemma 8.6 the  $A'$  entries sum to  $2 - E \leq 1$  at the start of step  $p + 1$ , so  $v_{p+1}(x) \leq 1$ ; and  $x$  is not deleted in steps  $t, \dots, p$  (all those heads are  $B_{\text{hi}}$ ), so it loses at most one per step and  $v_{p+1}(x) \geq v_t(x) - (p - t + 1)$ . Therefore  $\tau - t + 1 \leq 1 + (p - t + 1)$ , i.e.  $\tau \leq p + 1 = \tau - L + 1$ , i.e.  $L \leq 1$ .  $\square$

**Lemma 8.8** (backward induction). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-9] In  $\text{Fan}(\tau, L)$  under the reductio hypothesis, assume  $E = 0$ . Then the  $A'$  residue at the start of step  $p + 1$  is  $1 + 1$ , not a single 2.*

*Proof.* Write  $M_j$  for the  $A'$  value multiset at the start of step  $j$ . Since all of  $B_{\text{hi}}$  (Lemma 4.5(b)) and all of  $C$  (hypothesis) lie in  $\text{block}_j$ , the number of  $A'$  entries in  $\text{block}_j$  is  $a_j = D_j - (p - j) - (L + 1) = g_j - \tau \geq 1$ , and — the block being a prefix of the sorted list — they are the  $a_j$  largest entries of  $M_j$ . In particular a maximum entry of  $M_j$  is decremented at every step. By Lemma 8.4 the total of  $M_{p+1}$  is 2, so it is  $1 + 1$  or a single 2; suppose it is a single 2. Say  $M_{j+1}$  has a unique maximum  $v$  and does not contain the value  $v - 1$ . At step  $j$  let  $M_j$  have maximum  $w$  of multiplicity  $\mu$ . If  $a_j < \mu$  the maximum does not drop,  $w = v$ , and the  $a_j \geq 1$  decremented copies land at  $v - 1$ , which  $M_{j+1}$  does not contain — a contradiction. So  $a_j \geq \mu$ : all maxima drop,  $w = v + 1$ , and  $M_{j+1}$  contains  $\mu$  copies of  $v$ , forcing  $\mu = 1$ . Every other entry of  $M_j$  is either in the block (its  $M_{j+1}$  value plus 1) or out of it (unchanged), so all are at most  $\max(\text{other entries of } M_{j+1}) + 1$ . Hence the hypothesis propagates backwards:  $M_{p+1} = [2]$  has unique maximum 2 and no 1, so  $M_p = [3, 1^b]$ ,  $M_{p-1}$  has unique maximum 4 with the others  $\leq 2$ , and inductively  $M_{p-t}$  has maximum  $3 + t$  with all other entries  $\leq t + 1$  — in particular  $v - 1$  is never present, so the induction never stops. At  $t = p - 1$  this gives  $\max(M_1) = p + 2$ . But  $M_1$  is the multiset of  $A'$  degrees, and every  $a \in A'$  has  $N(a) \subseteq B_{\text{hi}}$ , so  $\max(M_1) \leq p$ .  $\square$

The induction does not apply to  $1 + 1$ : its maximum is not unique — the value 1 with multiplicity 2 — and the first step needs a unique maximum. That is exactly why the lemma kills only the single-2 case, and it is not a gap. (An earlier text gave a different and false reason, namely that the value  $v - 1 = 0$  is present among the zero entries; that fails whenever  $|A'| = 2$ , which is realisable at every  $L \geq 2$ .)

**Theorem 8.9** (the fan family is eliminated). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-10] For every  $\tau$  and every  $L \geq 2$  there is no graph carrying the configuration  $\text{Fan}(\tau, L)$  and satisfying  $\text{residue}(G) = \alpha(G)$ . Consequently, by Fact 2.2, every  $\text{Fan}(\tau, L \geq 2)$  graph has residue  $\leq \alpha - 1$ .*

*Proof.* Assume residue  $= \alpha$ , i.e.  $s = \tau$ . By Lemma 8.2 the heads of steps  $1, \dots, p$  are exactly  $B_{\text{hi}}$ . By Lemma 8.7,  $E = 0$ , so by Lemma 8.4 the  $A'$  entries sum to exactly 2 at the start of step  $p + 1$ , and by Lemma 8.8 they are  $1 + 1$ ; the  $C$ -entries are then all at  $\tau - p = L$ . So the list at the start of step  $p + 1$  is  $[L]^{L+1}, 1, 1$  (plus zeros), and Lemma 8.3 says  $L + 1$  further steps are needed, giving  $s = \tau + 1$  — a contradiction.  $\square$

The conclusion is stated as  $\text{residue} \leq \alpha - 1$  and not as  $s = \tau + 1$ . The stronger reading is available only *inside* the refuted hypothesis and may not be carried out of it; an earlier version of this theorem carried it out, which a review round caught. What is true, and measured, is stated separately:

**Observation 8.10.** *[computational: 1 098 141 strict and 1 700 094 superset  $\text{Fan}(\tau, L \geq 2)$  degree sequences in a box  $\tau \leq 10, R \leq 20, |A'| \leq 14]$  In every one of those instances  $\text{residue} = \alpha - 1$  and  $s = \tau + 1$ . This is a lead, not a theorem: at  $s = \tau + 1$  the reductio-only toolkit (Lemmas 4.1 and 4.5(b)) is unavailable — witness  $\text{Fan}(4, 2)$  with  $A' = [2, 1, 1]$ ,  $\deg = [5, 5, 4, 4, 4, 2, 1, 1]$ , where every  $B_{\text{hi}}$  degree is  $\tau + 1 = 5 = s$ . Separately, over 508 239 labelled high-phase runs under adversarial tie-breaks,  $E = 0$  and the residue partition is  $(1, 1)$  in all 508 239; and over 450 303 superset sequences the multiset at the start of step  $p + 1$  is exactly  $[L]^{L+1}$  with  $A'$  residue  $(1, 1)$  every time, the partition  $(2)$  never occurring. These runs are mechanism checks off the reductio hypothesis in the sense of Section 1.3; the lemmas rest on their proofs alone.*

**Corollary 8.11** (the hard core has  $L \geq 3$ ). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-19] In the hard core the number of  $B$ -vertices of degree  $\leq \tau$  satisfies  $L \geq 3$ ; together with Theorem 6.2, the residual hard core has  $\tau \geq 4$  and  $L \geq 3$ .*

*Proof.*  $L = 0$  is Theorem 4.6 with Observation 5.2;  $L = 1$  is Corollary 7.7. For  $L = 2$ : by Proposition 7.11 every hard-core instance with  $L = 2$  is rigidly  $\text{Fan}(\tau, 2)$  — clauses (a)–(d) supply every clause of Definition 8.1, the clause on  $A'$  coming from  $\deg_A(b_i) = 1$  — and Theorem 8.9 says no such graph satisfies  $\text{residue} = \alpha$ .  $\square$

The asymmetry here is deliberate. Theorem 8.9 kills the whole family  $\text{Fan}(\tau, L)$  for every  $L \geq 2$ , but it removes only the  $L = 2$  layer of the hard core, because rigidity is available at  $L = 2$  only. For  $L \geq 3$  a hard-core instance need not be a fan; Section 9 supplies the rigidity that is available.

## 8.1 A tail lemma

**Lemma 8.12** (tail). *[certified: two review rounds, distinct model families, zero mathematics defects; row R-15] Let the multiset at the start of some step be  $[L]^{L+1} \cup \lambda$  with  $\lambda$  a non-empty partition,  $\lambda_1 := \max \lambda$ . If  $L \geq \lambda_1$  then the number of further Havel–Hakimi steps is  $(L - \lambda_1) + s_0(\lambda)$ , where  $s_0(\lambda) := \text{steps}([\lambda_1]^{\lambda_1+1} \cup \lambda)$ . Consequently, still under  $L \geq \lambda_1$ , “clears in exactly  $L$  steps” is equivalent to  $s_0(\lambda) = \lambda_1$ , a condition on  $\lambda$  alone.*

*Proof.* If  $L = \lambda_1$  the list already is  $[\lambda_1]^{\lambda_1+1} \cup \lambda$  and the count reads  $0 + s_0(\lambda)$ ; so assume  $L > \lambda_1$ . Suppose at some stage the list is  $[t]^{t+1} \cup \lambda$  with  $t > \lambda_1$ , which holds at  $t = L$ . The head is  $t$ ; the remaining entries of value  $t$  are exactly the other  $t$  copies, and every  $\lambda$ -entry is  $< t$ ; so the block — the top  $t$  non-head entries — is exactly those  $t$  copies, they become  $t - 1$ , and  $\lambda$  is untouched. Hence  $[t]^{t+1} \cup \lambda \rightarrow [t - 1]^t \cup \lambda$  in one step. Iterating from  $t = L$  down to  $t = \lambda_1$  uses  $L - \lambda_1$  steps and leaves  $[\lambda_1]^{\lambda_1+1} \cup \lambda$ .  $\square$

Three hypotheses in that statement are there because review rounds put them there:  $\lambda \neq \emptyset$ , because  $\lambda_1$  is undefined otherwise; the case  $L = \lambda_1$  treated separately, because the induction’s premise  $t > \lambda_1$  is false at  $t = L$  there; and the standing range  $L \geq \lambda_1$  attached to the “ $L$ -free” consequence, which is false without it —  $\lambda = [2]$ ,  $L = 0$  has  $s_0 = 2 = \lambda_1$  yet takes one step, and  $\lambda = [3]$ ,  $L = 1$  clears in  $L = 1$  step although  $s_0([3]) = 4 \neq 3$ . Every use of the lemma in this paper is in range.

*Remark 8.13.* [computational: 1817 pairs  $(\lambda, L)$  with  $\lambda \vdash 2\nu$ ,  $\nu \leq 6$ , and  $\lambda_1 \leq L \leq 15$ ] The predicted count  $(L - \lambda_1) + s_0(\lambda)$  was compared against direct simulation on all of them, with 0 mismatches. Observation 8.5's second half was independently confirmed for  $L = 2, \dots, 9$  by direct simulation, and a reviewing session supplied a second, independent proof of it.

## 9 Rigidity of the $B$ -universal layer

Corollary 8.11 leaves  $L \geq 3$ , where no analogue of Proposition 7.11 was available. This section supplies one. It rests on a one-line degree cap, and it is the point at which the residual configuration space acquires a name.

**Lemma 9.1** (degree cap on low vertices). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-20*] *Let  $A$  be a maximum independent set,  $B = V \setminus A$ ,  $\tau = |B|$ , and let  $b \in B_{\text{lo}}$  have  $n_b$  non-neighbours in  $B$ . Then  $1 \leq \deg_A(b) \leq n_b + 1$ .*

*Proof.*  $\deg_A(b) \geq 1$ : otherwise  $N(b) \subseteq B$ , so  $A \cup \{b\}$  is independent, contradicting maximality of  $A$ . For the upper bound,  $\deg_B(b) = (\tau - 1) - n_b$ , and  $b \in B_{\text{lo}}$  means  $\deg(b) \leq \tau$ , so  $\deg_A(b) = \deg(b) - \deg_B(b) \leq \tau - (\tau - 1 - n_b) = n_b + 1$ .  $\square$

The hypotheses used are worth separating, and this is the one place in the campaign's report that advertises minimality of hypotheses, so a review round tested it there. The *lower* bound uses only that  $A$  is *maximal* — cardinality is never invoked — and the *upper* bound uses neither maximality nor maximum cardinality, only the definitions of  $B_{\text{lo}}$ ,  $\tau$  and  $n_b$ . So Lemma 9.1 holds verbatim for any maximal independent set. We state it with “maximum” because every consumer supplies one anyway and because  $\alpha = |A|$  is used in the same breath.

**Corollary 9.2.** [*certified: two review rounds, distinct model families, zero mathematics defects; row R-20*] *A  $B$ -universal low vertex has  $\deg_A(b) = 1$  exactly.*

*Proof.*  $n_b = 0$  in Lemma 9.1.  $\square$

**Theorem 9.3** (rigidity). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-21*] *Work in the hard-core frame and suppose  $B_{\text{lo}} \neq \emptyset$  and every low vertex is  $B$ -universal, i.e.  $B_{\text{lo}}^+ = \emptyset$ . Then:*

- (a)  $\deg_A(b) = 1$  for every  $b \in B_{\text{lo}}$ ;
- (b) the unique  $A$ -neighbours of the low vertices all coincide in one vertex  $a_0$ ;
- (c)  $a_0$  is adjacent to all of  $B$ ;
- (d) every non-edge of  $B$  lies inside  $B_{\text{hi}}$ , i.e.  $\nu = \bar{m}$ , and  $\nu \geq 1$ ;
- (e) every  $a \in A \setminus \{a_0\}$  has  $N(a) \cap B_{\text{lo}} = \emptyset$ .

Consequently  $G$  carries exactly the configuration  $\text{GFan}(\tau, L, \nu)$  of Definition 8.1, with  $\nu = \bar{m} \geq 1$ . In the hard core — that is, adding the *reductio hypothesis* — one has in addition  $\nu \leq L - 1$ .

*Proof.* (a) is Corollary 9.2. For (b), any two low vertices are adjacent by  $B$ -universality, so Lemma 5.1 gives them a common  $A$ -neighbour, which by (a) must be the unique  $A$ -neighbour of each; hence all coincide, in a vertex  $a_0$ . (For  $L = 1$  take  $a_0$  to be the unique  $A$ -neighbour of the unique low vertex.) For (c), fix  $b \in B_{\text{lo}}$ ; for every  $w \in B \setminus \{b\}$  we have  $b \sim w$ , so Lemma 5.1 gives a common  $A$ -neighbour of  $b$  and  $w$ , which by (a) is  $a_0$ ; hence  $a_0 \sim w$  for all  $w \in B \setminus \{b\}$ , and  $a_0 \sim b$  by definition. For (d), a non-edge of  $B$  incident to a low vertex would exhibit a non-neighbour

of that vertex in  $B$ , contradicting  $B$ -universality; so every non-edge lies inside  $B_{\text{hi}}$  and  $\nu = \bar{m}$ , and  $\nu \geq 1$  is Observation 5.2. For (e), if  $a \neq a_0$  had a neighbour  $b \in B_{\text{lo}}$  then  $\deg_A(b) \geq 2$ , contradicting (a).

The clauses of Definition 8.1 are exactly (a)+(b)+(c), (d), (e), together with “ $B_{\text{hi}}$  is high”, which is the definition of  $B_{\text{hi}}$ . Conversely,  $\text{GFan}(\tau, L, \nu)$  is the conjunction of exactly those clauses and asserts nothing else, so the match is an equality and not merely an implication — which is what “exactly” claims. The final bound  $\nu \leq L - 1$  is Corollary 7.8 applied to this hypothesis, and that corollary rests on Theorem 7.6 and hence Theorem 7.3, which live in the hard core.  $\square$

*Remark 9.4* (scope, stated exactly). Clauses (a)–(c) and (e) need only that  $A$  is a maximum independent set, plus Lemma 5.1; clause (d) needs additionally  $\text{diam} = 4$ , through Observation 5.2. So (a)–(e) hold in the hard-core frame, with no reductio hypothesis. Only the bound  $\nu \leq L - 1$  needs the hard core, and that is not a formality: see Remark 9.6. An earlier version of this paragraph glossed Lemma 5.1 as though its only hypothesis were  $f = \alpha + 1$ , and thereby claimed that (a)–(c),(e) survive outside the frame. *That gloss is not licensed.* Lemma 5.1 as certified is a statement in the frame, and nothing in the campaign’s work establishes a version of it assuming  $f = \alpha + 1$  alone. The gloss was an affirmative hypothesis-weakening claim about an import, which is exactly the class of claim the campaign’s scope discipline exists to police; a review round found it, and it is withdrawn here. Theorem 9.3’s own statement and proof are unaffected.

**Corollary 9.5** (the  $B$ -universal layer collapses to  $\nu \geq 2$ ). [*certified: two review rounds, distinct model families, zero mathematics defects; row R-21*] *In the hard core with  $L \geq 2$  and every low vertex  $B$ -universal, Theorem 9.3 gives  $\text{GFan}(\tau, L, \nu)$  with  $1 \leq \nu \leq L - 1$ ; the case  $\nu = 1$  is exactly  $\text{Fan}(\tau, L)$ , which Theorem 8.9 eliminates. Hence that layer reduces to  $\text{GFan}(\tau, L, \nu)$  with  $2 \leq \nu \leq L - 1$  — in particular it is empty for  $L = 2$ , and at  $L = 3$  it reduces to the single configuration  $\text{GFan}(\tau, 3, 2)$ .*

*Remark 9.6.* [computational: 2931 pairs (graph, maximum independent set) over the exhaustive atlas  $n \leq 7$ , 4382 over 1200 random connected graphs  $n = 8, \dots, 10$ , 1737 over 400 random connected  $n = 11, 12$ , and 622 targeted hard-core-frame instances] Lemma 9.1 and Corollary 9.2 were tested on 1926  $B$ -universal low vertices with  $\deg_A = 1$  in every one and 0 failures. Uniform random graphs almost never land in the frame — 12 frame instances across the 2195 exhaustive and random graphs — so a targeted generator was written: build  $B$  with a chosen non-edge set, attach  $A$ -vertices with random non-empty types, and keep only instances passing connected,  $\text{diam} = 4$ , non-forest,  $f = \alpha + 1$  and “ $A$  is a maximum independent set”. That yields 622 frame instances, of which 31 satisfy  $B_{\text{lo}}^+ = \emptyset$ ; all five conclusions of Theorem 9.3 hold in 31 of 31. The scope demarcation of Remark 9.4 is measured, not assumed: on the hypothesis set the  $(L, \nu)$  histogram is  $\{(1, 1) : 22, (1, 3) : 2, (2, 1) : 3, (3, 1) : 4\}$ , and the two instances with  $(L, \nu) = (1, 3)$  have  $\nu = 3 > L - 1 = 0$  — so the bound  $\nu \leq L - 1$  is *false* in the frame and genuinely needs the reductio hypothesis, exactly as the proof says. None of the 622 frame instances satisfies  $\text{residue} = \alpha$ .

## 10 Statements at one adversarial review round

Everything in this section is proved, and everything in it has passed exactly *one* review round — a Qwen3.8-Max session — returning no mathematics defect against these statements. The second round, which the campaign’s protocol requires from a different model family before a statement is certified, was dispatched and did not deliver: the brief cited four imports by name without stating them, so the reviewing session verified every downstream calculation *conditionally on the quoted properties* and correctly refused to certify the imports it could not see. A review round

obstructed by a defect in its own brief is not a round. These statements are therefore reported here in a separate tier, and the reader should treat them accordingly.

The setting is  $\text{GFan}(\tau, L, \nu)$  of Definition 8.1 under the *reductio hypothesis*;  $C = B_{\text{lo}} \cup \{a_0\}$ ,  $p = \tau - L = |B_{\text{hi}}|$ ,  $R = \sum_{a \in A'} \deg(a)$ , and  $E$  is the escape count of the high phase as in Section 8. The set-up carries over verbatim from there: every  $c \in C$  has  $\deg(c) = \tau$  exactly, every  $w \in B_{\text{hi}}$  has  $\deg(w) \geq \tau + 1$ , every  $a \in A'$  has  $\deg(a) \leq p < \tau$ , and so Lemma 8.2 applies word for word — its proof uses only that the  $B_{\text{hi}}$  vertices are the vertices of degree  $\geq \tau + 1$ , and Lemma 4.5(c) for the rest.

**Lemma 10.1** (residue mass, general  $\nu$ ). [*one round: Qwen3.8-Max; second family pending*] In  $\text{GFan}(\tau, L, \nu)$  under the *reductio hypothesis*, at the start of step  $p + 1$  the  $A'$  entries sum to exactly  $2\nu - E$ , and the  $C$ -entries are  $L + e_c$  for  $c \in C$ , where  $\sum_c e_c = E$ .

*Proof.* Exactly the count of Lemma 8.4 with  $\nu$  in place of 1. By Lemma 8.2 and Lemma 4.5(b),  $\sum_{j \leq p} D_j = \sum_{B_{\text{hi}}} \deg - \binom{p}{2}$ . Here  $\sum_{x \in B_{\text{hi}}} \deg_B(x) = p(\tau - 1) - 2\nu$ , since each of the  $\nu$  non-edges lies inside  $B_{\text{hi}}$  and is missed by both endpoints, and  $\sum_{x \in B_{\text{hi}}} \deg_A(x) = p + R$ ; so  $\sum_{B_{\text{hi}}} \deg = p\tau - 2\nu + R$ . Splitting the decrements by recipient —  $\binom{p}{2}$  to later high heads,  $p(L + 1) - E$  to  $C$ , the rest to  $A'$  — and using  $p(L + 1) = p\tau - p^2 + p$  and  $2\binom{p}{2} = p^2 - p$  gives  $\text{dec}_{A'} = R - 2\nu + E$ , so the  $A'$  total is  $R - \text{dec}_{A'} = 2\nu - E$ . Each  $c \in C$  starts at  $\tau$  and is decremented  $p - e_c$  times, ending at  $L + e_c$ .  $\square$

**Lemma 10.2** (escape bound, general  $\nu$ ). [*one round: Qwen3.8-Max; second family pending*] In  $\text{GFan}(\tau, L, \nu)$  under the *reductio hypothesis*, if  $E \geq 1$  then  $L \leq 2\nu - E$ .

*Proof.* Exactly the argument of Lemma 8.7. At an escape step  $t$  with  $e_t \geq 1$  escapes, the head is high, so  $D_t \geq \tau - t + 2$ ; the  $p - t$  remaining high vertices lie in  $\text{block}_t$ , and so do the  $(L + 1) - e_t$  non-escaping  $C$ -vertices; hence  $|\text{block}_t \cap A'| \geq (\tau - t + 2) - (p - t) - (L + 1) + e_t = 1 + e_t \geq 2$ , using  $\tau - p - L = 0$ . Pick  $x \in \text{block}_t \cap A'$ ; as  $\text{block}_t$  is a prefix and the escaping  $c \notin \text{block}_t$ ,  $v_t(x) \geq v_t(c) \geq \tau - t + 1$ . By Lemma 10.1 the  $A'$  mass at step  $p + 1$  is  $2\nu - E$ , so  $v_{p+1}(x) \leq 2\nu - E$ ; and  $x$  is not deleted in steps  $t, \dots, p$ , so  $v_{p+1}(x) \geq v_t(x) - (p - t + 1)$ . Therefore  $\tau - t + 1 \leq (2\nu - E) + (p - t + 1)$ , i.e.  $L = \tau - p \leq 2\nu - E$ .  $\square$

**Lemma 10.3** (backward induction, general form). [*one round: Qwen3.8-Max; second family pending*] In  $\text{GFan}(\tau, L, \nu)$  under the *reductio hypothesis*, the  $A'$  multiset at the start of step  $p + 1$  cannot have a unique maximum  $w \geq 1$  whose second-largest entry is  $\leq w - 2$ .

*Proof.* As in Lemma 8.8, with the block count  $a_j = D_j - (p - j) - (L + 1) + e_j \geq 1 + e_j \geq 1$ , so that a maximum entry of  $M_j$  is decremented at every step. Suppose  $M_{j+1}$  has a unique maximum  $v$  and no entry of value  $v - 1$ ; the multiplicity argument of Lemma 8.8 shows  $\mu = 1$  and that every other entry of  $M_j$  is at most  $\max(\text{other entries of } M_{j+1}) + 1$ . Hence, writing  $\sigma_j$  for the second-largest entry of  $M_j$ , the pair (unique maximum  $v_j$ ,  $\sigma_j \leq v_j - 2$ ) propagates backwards: from  $(w, \sigma \leq w - 2)$  at  $p + 1$  one gets  $v_{p+1-t} = w + t$  and  $\sigma_{p+1-t} \leq \sigma + t \leq v_{p+1-t} - 2$  for every  $t$ , so the induction never stops. At  $t = p$  it gives  $\max(M_1) = w + p$ , while  $M_1$  is the multiset of  $A'$  degrees and every  $a \in A'$  has  $N(a) \subseteq B_{\text{hi}}$ , so  $\max(M_1) \leq p$ .  $\square$

Two conventions attach to Lemma 10.3 and both were fixed by review findings. First, for the purpose of the phrase “second-largest entry”, a residue with a single positive part is read as having second-largest 0, whether or not a real zero entry is present; the lemma’s operative content is “there is no entry equal to  $w - 1$ ”, and for a single-part residue  $[w]$  with  $w \geq 2$  that is true under either reading, while  $[1]$  is excluded under either. Second, the residues the lemma kills are

exactly those with a gap of at least 2 below a unique top:  $[w]$  for every  $w \geq 2$ ,  $[3, 1]$ ,  $[4, 1]$ ,  $[4, 2]$ , and so on; it does *not* apply to  $[1]$ ,  $[1, 1]$ ,  $[2, 1]$ ,  $[2, 2]$ ,  $[2, 1, 1]$ ,  $[1, 1, 1, 1]$ . An earlier text wrote “ $[w]$  for every  $w \geq 1$ ”, which is false at  $w = 1$  under the stated convention and undefined without it. No user is lost: every residue this paper feeds to the lemma is  $[2\nu]$  with  $2\nu \geq 2$ , or one of the survivors tabulated in Appendix C, all of which have  $w \geq 3$ .

**Theorem 10.4** (the  $\nu = 2$  layer). *[one round: Qwen3.8-Max; second family pending] For every  $\tau$  and every  $L \geq 3$  there is no graph carrying  $\text{GFan}(\tau, L, 2)$  and satisfying  $\text{residue}(G) = \alpha(G)$ .*

*Proof.* Assume  $\text{residue} = \alpha$ . By Lemma 8.2 the heads of steps  $1, \dots, p$  are  $B_{\text{hi}}$ , so exactly  $L$  steps remain after step  $p$ ; the multiset at the start of step  $p + 1$  is  $\{L + e_c : c \in C\}$  together with an  $A'$  residue of total  $2\nu - E = 4 - E$  (Lemma 10.1), and it must clear in exactly  $L$  further steps.

*Step 1: the escape budget.* By Lemma 10.2,  $E \geq 1$  forces  $L \leq 4 - E$ . So  $L \geq 4$  gives  $E = 0$ , and  $L = 3$  gives  $E \leq 1$ .

*Step 2:  $L \geq 4$ .* Here  $E = 0$ , the  $C$ -part is  $[L]^{L+1}$  and the  $A'$  residue is a partition of 4. The partitions  $[4]$  and  $[3, 1]$  have a unique maximum with the next entry at least 2 below it, so Lemma 10.3 kills both. For the remaining three, run the process: while the common  $C$ -value is  $t \geq 3$  the head is a  $C$ -entry, the other  $C$ -entries are the only entries of value  $\geq 3$ , so the block is exactly those and the  $A'$  entries (all  $\leq 2$ ) are untouched; hence after  $L - 2$  steps the list is  $[2]^3$  together with the untouched residue. Then  $[2, 2]$ :  $[2]^5 \rightarrow [2, 2, 1, 1] \rightarrow [1, 1, 0] \rightarrow 0$ , three further steps, total  $L + 1$ ;  $[2, 1, 1]$ :  $[2, 2, 2, 2, 1, 1] \rightarrow [2, 1, 1, 1, 1] \rightarrow [1, 1, 0, 0] \rightarrow 0$ , three further, total  $L + 1$ ;  $[1, 1, 1, 1]$ :  $[2, 2, 2, 1, 1, 1, 1] \rightarrow [1]^6 \rightarrow \dots$ , four further, total  $L + 2$ . All exceed  $L$ , so none can occur.

*Step 3:  $L = 3$ .* The complete list of admissible shapes is eight rows ( $E \in \{0, 1\}$ ;  $E \geq 2$  is excluded by Step 1):

| $E$ | $C$ -part      | $A'$ residue   | steps to clear                  |
|-----|----------------|----------------|---------------------------------|
| 0   | $[3, 3, 3, 3]$ | $[4]$          | <b>3</b> , killed by Lemma 10.3 |
| 0   | $[3, 3, 3, 3]$ | $[3, 1]$       | $4 \neq 3$ , and also killed    |
| 0   | $[3, 3, 3, 3]$ | $[2, 2]$       | $4 \neq 3$                      |
| 0   | $[3, 3, 3, 3]$ | $[2, 1, 1]$    | $4 \neq 3$                      |
| 0   | $[3, 3, 3, 3]$ | $[1, 1, 1, 1]$ | $5 \neq 3$                      |
| 1   | $[4, 3, 3, 3]$ | $[3]$          | <b>3</b> , killed by Lemma 10.3 |
| 1   | $[4, 3, 3, 3]$ | $[2, 1]$       | $4 \neq 3$                      |
| 1   | $[4, 3, 3, 3]$ | $[1, 1, 1]$    | $4 \neq 3$                      |

Only the two single-part residues clear in exactly  $L = 3$  steps, and Lemma 10.3 kills both. Hence no case survives.  $\square$

**Corollary 10.5.** *[one round: Qwen3.8-Max; second family pending] In the hard core with  $L = 2$ , Proposition 7.11(a),(b) force  $B_{10}^+ = \emptyset$ ; Theorem 9.3 then gives Proposition 7.11(c),(d) directly, and  $\nu \leq L - 1 = 1$  with  $\nu \geq 1$  gives  $\nu = \bar{m} = 1$ , i.e.  $\text{Fan}(\tau, 2)$ .*

This removes the tightness re-run from Proposition 7.11(c):  $\deg_A(b) = 1$  follows from the degree cap alone. The campaign had pre-registered “tightness arguments need the inequality chain to be tight at every link, which is not spelled out” as a likely failure mode of Proposition 7.11 before the review round that examined it; the corollary is what makes the concern moot.

**Corollary 10.6.** *[one round: Qwen3.8-Max; second family pending] In the hard core, if every low vertex is  $B$ -universal then  $L \geq 4$  and  $\nu \geq 3$ .*

*Proof.* Theorem 9.3 makes the instance  $\text{GFan}(\tau, L, \nu)$  with  $1 \leq \nu \leq L - 1$ ;  $\nu = 1$  is  $\text{Fan}(\tau, L)$ , killed for  $L \geq 2$  by Theorem 8.9, and  $\nu = 2$  is killed for  $L \geq 3$  by Theorem 10.4.  $L = 1$  cannot occur, since  $1 \leq \nu \leq L - 1$  already forces  $L \geq 2$ ; and  $L = 2$  forces  $\nu = 1$ . So  $\nu \geq 3$ , whence  $L \geq \nu + 1 \geq 4$ .  $\square$

**Corollary 10.7.** [*one round: Qwen3.8-Max; second family pending*] *In the hard core with  $L = 3$ , some low vertex is not  $B$ -universal ( $B_{\text{lo}}^+ \neq \emptyset$ ).*

*Remark 10.8* (the other layer is not excluded at  $L \geq 3$ ). At  $L = 2$ , Proposition 7.11(a),(b) forced  $B_{\text{lo}}^+ = \emptyset$  via Theorem 7.6 and Fact 5.3. That argument does not survive  $L = 3$ , and we record why so that no successor re-derives it as a theorem. With  $k := |B_{\text{lo}}^+| = 1$  and  $\nu(B_{\text{lo}}) = 0$ , Theorem 7.6 reads  $1 + c + \bar{m} \leq L - 1 = 2$ , which is satisfiable in two ways:  $c = 0, \bar{m} = 1$ , where  $T_1, T_2 \subseteq B_{\text{hi}}$  are singletons spanning the unique  $B_{\text{hi}}$  non-edge and no contradiction with Fact 5.3 arises because  $\bar{m} \neq 0$ ; and  $c = 1, \bar{m} = 0$ , where the non-universal low vertex is itself in  $T_1$  and  $T_2$  sits inside its  $B_{\text{hi}}$  non-neighbourhood, so Lemma 5.4 gives  $\deg_A \geq 3$  there and Lemma 9.1 forces  $n_b \geq 2$  for that vertex. Both are budget-consistent. The contradiction that closed  $L = 2$  used  $\bar{m} = 0$  and  $c = 0$  simultaneously, and at  $L \geq 3$  the budget no longer forces both. So the residual hard core splits as

$$L \geq 3 = [B\text{-universal layer: } \text{GFan}(\tau, L, \nu), 2 \leq \nu \leq L - 1] \cup [B_{\text{lo}}^+ \neq \emptyset].$$

Lemma 9.1 is the handle on the second layer:  $\deg_A(b) \leq n_b + 1$  converts every  $A$ -degree lower bound into a non-edge lower bound. Concretely, Lemma 5.4 now reads: every low vertex in  $T_1 \cup T_2$  has at least two non-neighbours in  $B$ ; and Lemma 5.1's contribution to Theorem 7.6 is recovered by Lemma 9.1 for free.

## 11 A computer-assisted elimination, and a false quantifier

This section reports one theorem at the weakest tier used in this paper, and it reports it together with the reason it sits there: the previous statement at this position was *false*, a review round produced a counterexample family, and the repaired statement has not yet been reviewed. We consider the episode worth reporting in full rather than silently repairing, because a pipeline that publishes only its successes gives no evidence about its failure modes, and because the defect is of a species — a quantifier admitting a degenerate instance nobody constructed — that no amount of numerical checking of the non-degenerate range would have found.

### 11.1 The statement, and the case list

**Theorem 11.1.** [*no completed round: the statement below is the repaired one; the text that was reviewed was false, and the repaired text is unreviewed*] *For every  $\tau$ , every  $\nu$  with  $1 \leq \nu \leq 6$ , and every  $L \geq \nu + 1$ , there is no graph carrying the configuration  $\text{GFan}(\tau, L, \nu)$  and satisfying  $\text{residue}(G) = \alpha(G)$ .*

*Proof.* Four proved bounds make the case list finite for each  $\nu$ : Lemma 10.1 (the  $A'$  residue totals  $2\nu - E$  and the  $C$ -part is  $\{L + e_c\}$  with  $\sum e_c = E$ ); Lemma 10.2 ( $E \geq 1 \Rightarrow L \leq 2\nu - E$ , so  $E \geq 1$  occurs only for the finitely many pairs  $\nu + 1 \leq L \leq 2\nu - E$ , and  $E \leq \nu - 1$ ); Corollary 7.8 ( $L \geq \nu + 1$ ); and Lemma 8.12 (for  $E = 0$  and  $L \geq \lambda_1$ , survival depends on  $\lambda$  alone, and the finitely many boundary rows  $\nu + 1 \leq L < \lambda_1 \leq 2\nu$  are checked directly). Every shape in the resulting finite list either fails to clear in exactly  $L$  steps or is killed by Lemma 10.3. The enumeration is reproduced in full in Appendix C; the counts are

| $\nu$ | $E = 0$ rows Lemma 10.3 misses | $E \geq 1$ shapes tested | $E \geq 1$ rows it misses |
|-------|--------------------------------|--------------------------|---------------------------|
| 1     | none                           | 0                        | none                      |
| 2     | none                           | 3                        | none                      |
| 3     | none                           | 24                       | none                      |
| 4     | none                           | 110                      | none                      |
| 5     | none                           | 397                      | none                      |
| 6     | none                           | 1 211                    | none                      |

At  $E = 0$  and  $L \geq \lambda_1$  — inside Lemma 8.12’s range, where “clears in exactly  $L$  steps” is the  $L$ -free criterion  $s_0(\lambda) = \lambda_1$  — the only residue clearing in exactly  $L$  steps is the single part  $[2\nu]$ , for every  $1 \leq \nu \leq 6$ , and  $[2\nu]$  is killed by Lemma 10.3. Outside that range the finitely many boundary rows  $\nu + 1 \leq L < \lambda_1$  have their own survivors, tabulated in Appendix C, each with its own certificate.  $\square$

*Remark 11.2* (the demarcation, stated exactly). This is a computer-assisted proof and it is stated as such. The case list is finite *because* of Lemmas 10.1, 10.2, 8.12 and Corollary 7.8, all hand-proved. Within that finite list, Appendix C prints the  $E = 0$  criterion for all 159 partitions of  $2\nu$ ,  $\nu \leq 6$ ; all 51  $E = 0$  boundary rows; the closed form  $S(\nu) = \sum_{E=1}^{\nu-1} (\nu - E) p(E) p(2\nu - E)$  reproducing the “shapes tested” column; and the complete roster of the 72 surviving  $E \geq 1$  rows with their certificates. Every step except the *completeness of the  $E \geq 1$  survivor roster* is therefore checkable by hand from the text alone; that one step is a seconds-scale machine check over an explicitly specified, closed-form-counted finite set.  $\nu = 1$  is Theorem 8.9 and  $\nu = 2$  is Theorem 10.4, both with complete hand proofs;  $\nu = 3, \dots, 6$  rest on the enumeration.

An earlier version of the summary sentence in the proof read “at  $E = 0$  the only residue clearing in exactly  $L$  steps is the single part  $[2\nu]$ ”, with no range qualifier, and read literally that is contradicted by the paper’s own data:  $(L, \lambda) = (5, [7, 1])$  at  $\nu = 4$ ,  $(7, [9, 1])$  at  $\nu = 5$  and  $(7, [10, 1, 1])$  at  $\nu = 6$  are all  $E = 0$ , all clear in exactly  $L$  steps, and none is  $[2\nu]$ . They are boundary rows outside Lemma 8.12’s range, handled separately, and all are killed by Lemma 10.3; the conclusion is unaffected, but the sentence as written was false, and it was the punchline sentence. A review round found it, and the qualifier above is the repair.

**Corollary 11.3.** *[no completed round: inherits Theorem 11.1’s state] In the hard core, if every low vertex is  $B$ -universal then  $\nu \geq 7$  and  $L \geq 8$ .*

*Proof.* Theorem 9.3 gives  $\text{GFan}(\tau, L, \nu)$  with  $1 \leq \nu \leq L - 1$  — the lower bound is clause (d), i.e. Observation 5.2 — so  $L \geq \nu + 1$ ; Theorem 11.1 eliminates every  $\nu$  with  $1 \leq \nu \leq 6$ , which by that lower bound is every  $\nu \leq 6$  available here.  $\square$

## 11.2 The refutation

Theorem 11.1’s predecessor read “for every  $\nu \leq 6$ ”. A review round carried out by an OpenAI GPT-5.6 session through the `codex` interface returned the verdict `REFUTED`, with the following observation:  $\nu$  counts non-edges of  $G[B]$ , so  $\nu = 0$  is an admissible value, and the statement is false there.

**Proposition 11.4** (the counterexample family). *[computational:  $K_3, K_4, \dots, K_8$  all verified, every clause of Definition 8.1 checked individually, by two independent implementations] Every complete graph  $K_n$  with  $n \geq 3$  carries the configuration  $\text{GFan}(n - 1, n - 1, 0)$  and satisfies  $\text{residue} = \alpha$ . Explicitly for  $K_5$ : it is connected and not a forest,  $\alpha = 1$  with  $A = \{a_0\}$  a single vertex,  $f = 2 = \alpha + 1$ , the*

degree sequence is  $[4, 4, 4, 4, 4]$  and  $\text{residue} = 1 = \alpha$ ;  $\tau = 4$ ,  $B_{\text{lo}} = B$ ,  $L = 4$ ,  $B_{\text{hi}} = \emptyset$ ,  $\nu = 0$ , and  $L \geq \nu + 1$ .

The proof of Theorem 11.1 could never have covered  $\nu = 0$ , and inspecting why is instructive. At  $\nu = 0$ , Lemma 10.1 forces  $E = 0$  and makes  $\lambda$  the *empty* partition of 0; Lemma 8.12 is stated for  $\lambda$  non-empty; the boundary branch needs  $\lambda_1 \leq 2\nu = 0$ ; and the  $E \geq 1$  branch needs  $E \leq \nu - 1 = -1$ . All three branches of the case split are vacuous at  $\nu = 0$ , which is exactly why the enumeration in Appendix C starts at  $\nu = 1$  — the printed data already showed the gap, and nobody read it that way. The defect is a genuine gap between statement and proof, not a rider; the repair is the missing quantifier  $1 \leq \nu$ , and nothing downstream changes, because every consumer reaches the theorem through Theorem 9.3(d), where Observation 5.2 supplies  $\nu \geq 1$ . Corollary 11.3 is untouched for the same reason.

Three further things about the episode are worth stating, because they are the data a reader should use to calibrate how much this pipeline’s markers are worth.

*First, the round was suspected of being shallow before it was read, and the suspicion was wrong.* It ran faster and returned a smaller report than the preceding round, on a brief carrying more named joints; the campaign recorded that as a prior for shallowness. Measured against brief size the ranking inverts: the brief was 43 KB against the preceding round’s 84 KB and an earlier round’s 138 KB, runtime tracked brief size almost linearly, and per byte of brief this round produced more report than its predecessor. The lesson recorded in the campaign’s ledger is that predicting a reviewer’s depth from a surface statistic instead of from its work had by then failed three times.

*Second, the round did substantially more than referee.* It built its own graph rather than assuming the conclusion of Theorem 9.3: an eight-vertex frame instance with  $\alpha = 4$ ,  $\text{diam} = 4$ ,  $f = 5 = \alpha + 1$ , connected, not a forest,  $B_{\text{lo}} = \{b_0, b_3\}$  both  $B$ -universal,  $B_{\text{hi}} = \{b_1, b_2\}$ , unique  $B$  non-edge  $b_1b_2$  so  $\nu = 1$ , all five conclusions of Theorem 9.3 true, and  $\text{residue} = 3 \neq 4 = \alpha$ . Every claim about it was reproduced independently. That last line is what makes it a control rather than a witness: the frame holds and the *reductio* hypothesis fails, so Lemmas 10.1, 10.2 and 10.3 are unavailable on it, and the report says so and does not use it as evidence for them. It also declined to launder vacuity, stating in its own words that the positive- $\nu$  computation enumerates *shapes*, not realized graphs, so that the absence of an unkilld shape is not empirical evidence that no such graph exists.

*Third, the campaign’s own gate arithmetic went against the campaign.* Reading the report’s own joint-level verdicts, one could argue that the repaired statement  $1 \leq \nu \leq 6$  was reviewed in substance, since the specification, data and kill joints all came back clean and one of them says “clean for  $1 \leq \nu \leq 6$ ”. That reading was put to the orchestrator and *rejected*, on the pre-registered rule that a clean round counts only for the text as reviewed, and that a round which refutes a statement is not that statement’s clean round. Theorem 11.1 is therefore recorded at zero rounds, not at one, and Section 12 names the single round that would close it.

*Remark 11.5* (the data were reproduced a third time). [computational: 282 printed values re-derived from the specification alone and diffed against this text as parsed from the file; 0 mismatches] Because a refuting round raises the question of what *else* is wrong, the whole of Appendix C was recomputed by an implementation written from scratch — its own Havel–Hakimi and residue primitives, its own independence-number and forest-number routines by exhaustive subset enumeration, its own partition generator, its own clause checker for Definition 8.1 — which reuses nothing from the reviewing session or its checker, and which parses this paper’s own data blocks and diffs against its re-computation, so that a transcription slip here would surface rather than be laundered. Agreement

was exact on: the 159 values of  $s_0(\lambda)$  and the fact that the printed partition set is exactly the set of all  $p(2\nu)$  partitions at each  $\nu$ ; the  $E = 0$  tail survivors [2], [4], [6], [8], [10], [12]; the boundary pair counts 0, 1, 3, 7, 14, 26; the boundary survivors and their certificates;  $S(\nu) = 0, 3, 24, 110, 397, 1211$  both by enumeration and by closed form; the  $E \geq 1$  survivor counts 0, 1, 4, 9, 20, 38, every row and every certificate, with no row printed here that the recomputation lacks and none the other way; 0 misses of Lemma 10.3 over all  $9 + 72 + 6$  surviving rows; the 1 817 tail-formula pairs; and the zero-padding inertness control, extended to 18 963 (list, padding) pairs with 0 padding-dependent step counts.

## 12 What remains

### 12.1 The residual configuration space

Collecting Theorems 6.1, 6.2, 8.9 and 9.3 with Corollaries 8.11 and 9.5, a counterexample to the diameter-four dichotomy (D4) — equivalently, a graph in the hard core of Definition 3.13 — must satisfy

$$\tau \geq 4, \quad L \geq 3,$$

and must lie in one of the two layers of Remark 10.8: either every low vertex is  $B$ -universal, in which case Theorem 9.3 pins the graph to  $\text{GFan}(\tau, L, \nu)$  with  $2 \leq \nu \leq L - 1$  (and, granting Theorem 11.1 at its current tier, with  $\nu \geq 7$  and  $L \geq 8$ ); or some low vertex is not  $B$ -universal, in which case no rigidity statement is available and Lemma 9.1 is the only handle so far — it converts every  $A$ -degree lower bound into a lower bound on the number of non-neighbours in  $B$ , which is what a hypothetical  $L$ -uniform argument would consume.

Nothing here bounds  $\tau$  or  $L$  from above, and there is a measured reason to expect no such bound: the family with  $B$  independent,  $c$  universal  $A$ -vertices and two private leaves per  $B$ -vertex lands genuinely inside the hard-core frame for every  $\tau \geq 3$  and  $c \geq \tau$ , so a case analysis proceeding one  $\tau$  at a time can never terminate. That observation is why every statement in Sections 4–9 is  $\tau$ -uniform.

### 12.2 Two combinatorial conjectures that would close the $B$ -universal layer

The table underlying Theorem 11.1 is uniform in  $\nu$  as far as it was computed: at  $E = 0$  only  $\lambda = [2\nu]$  survives, and every  $E \geq 1$  survivor has a unique maximum with a gap of at least 2 below it. If that persists for all  $\nu$ , the entire  $B$ -universal layer of the hard core is empty at every  $L$ , and by Remark 10.8 the whole hard core reduces to the layer  $B_{10}^+ \neq \emptyset$ . Precisely:

**Conjecture 12.1.** [conjecture; verified for every  $\nu \leq 10$ ; proved for  $\lambda = [2\nu]$  and for every  $\lambda$  with at most two parts] For every  $\nu$  and every partition  $\lambda$  of  $2\nu$  other than  $[2\nu]$ ,  $s_0(\lambda) \neq \lambda_1$ .

**Conjecture 12.2.** [conjecture; verified for every  $\nu \leq 10$ ] Every  $(L, E)$  survivor with  $E \geq 1$  has an  $A'$  residue  $\lambda$  with a unique maximum whose second-largest entry is at most  $\lambda_1 - 2$ .

*Remark 12.3* (partial progress on Conjecture 12.1). [**no completed round**: hand proofs; never queued for review] Four statements were proved by hand and are recorded here at the weakest tier, having never been submitted to any review round. (i)  $s_0([2\nu]) = 2\nu$  for every  $\nu$ , by a two-step parity induction with no computation — so the  $[2\nu]$  row of the enumeration is hand-proved at every  $\nu$ , not only for  $\nu \leq 6$ . (ii) Conjecture 12.1 is not a step-count statement but a residue statement: writing  $k$  for the number of parts of  $\lambda$ , one has  $s_0(\lambda) = \lambda_1$  if and only if  $\text{residue}([\lambda_1]^{\lambda_1+1} \cup \lambda) = k + 1$ . (iii)

Consequently, by Fact 2.2, Conjecture 12.1 holds for a given  $\lambda$  as soon as  $[\lambda_1]^{\lambda_1+1} \cup \lambda$  admits a realization with independence number at most  $k$  — so the general case is reduced to a construction problem. (iv) For  $k \leq 2$  the construction succeeds:  $s_0((w, c)) = w + 1$  exactly, whenever  $w + c$  is even, which it always is here since  $\lambda \vdash 2\nu$ ; hence Conjecture 12.1 holds for every partition with at most two parts. The step counts were verified over all 10 268 terminating cases with  $n \leq 28$ , and the enumeration of Appendix C was separately extended to  $\nu \leq 10$  with the same outcome. That extension is deliberately *not* folded into Theorem 11.1’s statement: it was produced while review rounds on the  $\nu \leq 6$  text were in flight, and desynchronising a theorem from the text a reviewer is holding invalidates the round. When it is folded in it must be folded as  $1 \leq \nu \leq 10$ , or the hole of Section 11.2 re-opens.

### 12.3 The one review round that would close the certification surface

The mathematics of this line is, in the campaign’s judgement, complete as far as it goes; what is not complete is its certification. Exactly ten statements sit at one round (Section 10) and two at zero (Section 11), and a single correctly-briefed round by a non-Qwen family on the current repaired text would give the ten their second family and the two their first. The obstruction to the round that was attempted is documented above and is entirely on the dispatching side: the brief must *paste*, in full statement form, the four imports whose absence obstructed it — Lemma 8.2, which was not even in the assumable list; Lemma 4.5 clause (b) explicitly, rather than the lemma’s name; Proposition 7.11 clauses (a) and (b) verbatim; and Corollary 7.7, or better, drop its citation entirely, since Corollary 10.6 no longer uses it. That round was specified but not dispatched, the campaign having entered a wind-down.

## 13 Limitations

We state the limitations of this paper plainly.

1. *The conjecture is not resolved, and the gap is not small.* What is proved is a reduction and a narrowing: the diameter-four dichotomy (D4) implies the conjecture at diameter four, and the configurations that could violate (D4) have been narrowed to  $\tau \geq 4$ ,  $L \geq 3$  and two layers, one of which is pinned to an explicit family and one of which is not pinned at all. Nothing here bounds  $\tau$  or  $L$ , and Section 12 explains why no argument proceeding one  $\tau$  at a time can.
2. *Nothing is formalized.* The companion papers of this pipeline [7, 8] end in Lean 4 files whose compilation is the arbiter of every claim. This one does not, and the strongest marker used here is a statement about a review process. Two independent reviewing sessions from different model families returning no mathematics defect is materially weaker evidence than a kernel-checked proof, and the reader should treat it as such. The one thing the tiering does guarantee is that the paper distinguishes the levels rather than averaging them.
3. *The residual is at diameter four only in a measured range.* Corollary 3.6 settles  $d \in \{1, 2, 3, 5, 6, 9\}$  unconditionally;  $d = 4$ ,  $d = 7$ ,  $d = 8$  and every  $d \geq 10$  are not settled by the elementary package. What Remark 3.11 establishes is that in the exhaustively checked range  $n \leq 8$  every graph missed by the three certificates has  $d = 4$ . Proposition 3.12 is therefore a statement about the diameter-four case and not about “the residual case” in general, and the campaign’s own internal summary of it as the latter is an overstatement that we correct here.

4. *One certified proof is summarised, not reproduced.* Theorem 6.2’s complete case analysis — four branches, several sub-branches, and seven landed repairs from three review rounds — runs to several pages of dense case work in the source report, and Section 6 gives its skeleton and its tools rather than every branch. The two review rounds certified the full text, not this summary. A reader who wants the branch-level detail must go to the source report; a reader who wants to check Theorem 6.2 independently is better served by the parameterisation in Remark 6.3, which reduces the  $\tau = 3$  frame to an exhaustive scan over a box of multiplicities.
5. *Fact 2.2 is used as a black box and its bibliographic data comes from an automated pass.* The inequality residue  $\leq \alpha$  carries every argument in this paper. It is not reproved here, and the attribution should be re-checked against the literature before any formalization. It is asserted by the verification script on the whole exhaustive range and never fails there, which is evidence about the statement and none at all about the attribution.
6. *Numerical scopes are exactly what the markers say.* Every [computational: ...] marker names a finite range, and a failed bounded search inside it is not a proof of impossibility. In particular the hard core is empty on everything reachable at the sizes searched — which is what the conjecture predicts, and which therefore supplies no discriminating evidence about the unproved statements at all. The mechanism checks reported in Observation 8.10 and elsewhere run *off* the reductio hypothesis for exactly this reason, and they corroborate the shape of an argument rather than its conclusion.
7. *Two of the campaign’s own statements were false.* Remark 4.9 and Section 11.2 report them, with their counterexamples. Both were caught by the review gate, not by the solver. The honest reading of that is not that the gate works, but that the solver’s first-pass output should not be trusted without it; the success rate of the first gate, not of the pipeline, is the measure of a solver.
8. *Literature search was automated.* We checked for prior work on Conjecture 1.1 by automated search and found none beyond the collection and the formal corpus, but automated searches miss things. No priority is claimed for anything here, and readers should assume that some of the elementary material in Section 3 is known.
9. *Ten statements are one round short and two are two rounds short.* They are segregated into Sections 10 and 11 and marked accordingly. Corollary 11.3 in particular should not be quoted as established; the strongest unconditional consequence of the certified material is Corollary 8.11’s  $L \geq 3$ .

## Acknowledgement of process

No human contributed a mathematical step to this paper. The pipeline — problem selection, attack design, solving, adjudication, adversarial review, computation and drafting — was run autonomously. Responsibility for the correctness of what is claimed nevertheless rests where it always does: with the reader who checks it, and, for the statements below the top tier, with the reader who does not yet.

## A Certification ledger

Table 2 reproduces, for each statement of this paper carrying a *certified* or *one round* marker, the campaign’s registry row and the review artifacts behind it. “Family” records the model family of

each round; the protocol requires the two rounds of a certified statement to come from different families, and requires at least one of them to have read the final repaired text. The adjudication column names the campaign’s own reproduction script, which re-derives every claim of the report from scratch before it is upheld; those scripts are listed in Appendix B.

| row   | statements                                                      | review rounds (family)                                                                                                                                              | adjudication                                         |
|-------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| R-1   | Lemma 4.4                                                       | round A2 (Qwen3.8-Max) + round B (Claude Opus 5) + confirmation pass (Claude Opus 5, separate session)                                                              | w61_r4_a2check.py,<br>w61_thmK_check.py              |
| R-2   | Lemma 4.5                                                       | same chain                                                                                                                                                          | same                                                 |
| R-3   | Theorem 4.6                                                     | same chain                                                                                                                                                          | same                                                 |
| R-4   | Corollary 4.7                                                   | same chain                                                                                                                                                          | same                                                 |
| R-5   | Lemma 4.2                                                       | same chain                                                                                                                                                          | w61_adjudicate_r4.py                                 |
| R-6,  | the two superseded counting                                     | round B (Claude Opus 5) + con-                                                                                                                                      | w61_adjudicate.py                                    |
| R-7   | statements of Remark 4.9                                        | firmation pass                                                                                                                                                      |                                                      |
| R-9   | Lemmas 8.2, 8.3, 8.4, 8.8, 8.6, 8.7                             | rounds A, A2, B (Claude Opus 5, three separate sessions) + cross-family round (Qwen3.8-Max) + verification pass (Qwen3.8-Max)                                       | w61_r5_a2check.py,<br>w61_r6_q14check.py             |
| R-10  | Theorem 8.9                                                     | same chain                                                                                                                                                          | same                                                 |
| R-11, | Lemmas 4.1, 4.3 and the                                         | verification pass (Qwen3.8-                                                                                                                                         | w61_r5_generalS.py                                   |
| R-12  | general- <i>s</i> closure                                       | Max), joint V-M1                                                                                                                                                    |                                                      |
| R-13  | Theorem 6.2 with repairs R1, R3, J1, J2, K1, K2, K3             | round A (Claude Opus 5) + Part 2 of the verification pass (Qwen3.8-Max), both math joints re-derived independently                                                  | w61_r5_jcheck.py,<br>w61_tau3_struct.py              |
| R-14  | Observation 8.5                                                 | released by R-15; own content certified with R-9                                                                                                                    | w61_r5_bcheck.py                                     |
| R-15  | Lemma 8.12 with repairs Y1–Y3                                   | round 1 (Qwen3.8-Max, no mathematics defect) + round 2 ( <b>codex spark</b> , clean on the repaired text)                                                           | w61_r11_adjudicate.py,<br>w61_S3_TAIL_spark_check.py |
| R-16  | Theorem 7.2, Theorem 7.3, Lemma 7.1                             | round 1 (Qwen3.8-Max, no mathematics defect, repairs X1–X4) + round 2 ( <b>codex sol</b> , on the repaired text, Steps 1–6 of Theorem 7.3 re-derived independently) | w61_r10_adjudicate.py,<br>w61_r12_adjudicate.py      |
| R-17  | Theorem 7.6, Corollaries 7.7, 7.8, 7.10, Propositions 7.9, 7.11 | same two-family chain; all joints clean in round 2 with independent re-derivation, including the two-sided squeeze of Proposition 7.11(c)                           | same                                                 |
| R-18  | Corollary 7.5 with repair AA1                                   | same chain; AA1’s replacement argument certified directly by the orchestrator                                                                                       | w61_r12_adjudicate.py                                |
| R-19  | Corollary 8.11                                                  | released by R-16–R-18; own content certified with R-9/R-10                                                                                                          | —                                                    |

| row  | statements                                                   | review rounds (family)                                                                                                                             | adjudication                       |
|------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| R-20 | Lemma 9.1, Corollary 9.2 with repair AB3                     | round 1 (Qwen3.8-Max) + round 2 ( <code>codex sol</code> ; both bounds re-derived independently)                                                   | <code>w61_r13_adjudicate.py</code> |
| R-21 | Theorem 9.3, Corollary 9.5 with repair AB2                   | same two families; round 2 supplied an independently constructed frame witness (Section 11.2)                                                      | <code>w61_r13_adjudicate.py</code> |
| —    | Lemmas 10.1–10.3, Theorem 10.4, Corollaries 10.5, 10.6, 10.7 | <b>one round only:</b> Qwen3.8-Max, joints clean, repairs Z1–Z4 landed. The second family’s round was obstructed by a briefing defect (Section 10) | <code>w61_r12_adjudicate.py</code> |
| —    | Theorem 11.1, Corollary 11.3                                 | <b>no completed round:</b> the text reviewed was false (Section 11.2); the repaired text is unreviewed                                             | <code>w61_r13_adjudicate.py</code> |

Table 2: Certification ledger. Registry rows are the campaign’s own identifiers.

Two structural facts about this table are worth stating. First, the model families are genuinely separated: no reviewing session read another’s report, the campaign’s dispatch records, or the pre-declared failure modes, and each brief carried an explicit isolation rule. Second, five rows were held back at points where accepting a clean verdict wholesale would have promoted something unearned — Observation 8.5 was held until Lemma 8.12 cleared its own two-family bar, Corollary 8.11 was held until the whole of Section 7 cleared, and the entire Section 10 block was held and remains held. Those holds are the part of the ledger that carries information.

## B Verification-artifact inventory

Every script below is a stand-alone Python program in the campaign repository; none requires a solver, and the longest single run is about 100 seconds. Each begins by recomputing the two calibration values  $\text{residue}(K_2) = 1$  and  $\text{residue}(C_n) = \lceil n/3 \rceil$  for  $n = 3, \dots, 9$  from its own implementation, so that a mistranscription of the residue definition aborts the run before any reported number is produced. Paths are relative to the repository root.

### B.1 The falsification and certificate stages

#### `notes/proofs/wowii61_verify.py`

The primary verifier. Mode `all` runs stages A–E of Remark 3.10 and *asserts*  $\text{residue} \leq \alpha$  (Fact 2.2),  $f \geq \alpha + 1$  (Lemma 3.1),  $f \geq \alpha + \lfloor (d-1)/4 \rfloor + 1$  (Corollary 3.5) and (C61) itself on every graph it touches, so a counterexample to any of them aborts the run. Mode `w` emits the witnesses  $W_1, \dots, W_7$  of Section 3.7 with all their invariants. Mode `f` evaluates the two sub-conjectures of the campaign’s first round and the two dead routes of Section 3.7 over

46 995 graphs. It also produces the coverage table of Remark 3.11. `networkx` is used only for the atlas of graphs on at most seven vertices; everything else is bit manipulation.

## B.2 Toolkit and small- $\tau$

### `problems/wowii/w61_lemH.py`

Certifies Lemmas 4.1 and 4.2 and the head-decay bound they replaced, over 116 873 connected graphs ( $n \leq 7$  exhaustive, the  $n = 8$  cover, 12 000 random  $n = 9, \dots, 12$ , structured families), of which 36 612 satisfy the reductio hypothesis; canonical sort plus three randomised adversarial tie-breaks per graph.

### `problems/wowii/w61_blockocc.py`

Certifies Lemma 4.4 and the block-occupancy mechanism over 116 189 connected graphs, 36 664 of them under the reductio hypothesis, with the six labelled claims reported separately.

### `problems/wowii/w61_thmK_check.py`

Independent check of Theorem 4.6 (Remark 4.8), written by the owner agent and reusing no code from the agent that proposed the theorem: 114 914 connected graphs, 1 464 hypothesis hits, 0 non-cliques.

### `problems/wowii/w61_cstar.py`

Certifies Lemma 5.4 on the hard-core *frame* — no reductio hypothesis imposed — over the exhaustive  $n \leq 7$  atlas plus the  $n = 8$  cover: 340 frame graphs, 399 pairs realising Fact 5.3, 0 failures, plus the  $\tau$ -histogram of Remark 5.5.

### `problems/wowii/w61_famgen.py`

Generates the two unbounded hard-core-frame families used in Section 12 and confirms membership instance by instance for  $\tau = 3, \dots, 7$  and  $c = 2, \dots, 10$ .

### `problems/wowii/w61_tau3.py`

Exhaustive scan of the  $\tau = 3$  frame by the parameterisation of Remark 6.3: 1 119 744 parameter tuples in a multiplicity box of size 5, of which 591 710 lie in the frame; reports the  $e_B$  distribution and the minimum of  $m - \sum_{i \leq 3} D_i$ .

### `problems/wowii/w61_tau3_struct.py`

Certifies the structural skeleton of Theorem 6.2 — the three  $B$ -degree bounds and the two implications  $e_B = 1 \Rightarrow k \geq 1$ ,  $e_B = 2 \Rightarrow k \geq 2$  — on 147 348 instances, and runs the four explicit families of the proof at parameters up to  $X = 59$ .

### `problems/wowii/w61_r5_t3check.py`

Corroborating box for Theorem 6.2: 12 288 graphs with  $\tau = 3$  over all  $e_B$  and all type multiplicities  $\leq 2$ , with  $A$  verified maximum; 0 hard-core instances.

### `problems/wowii/w61_r3_repair.py`, `w61_r5_jcheck.py`

Numerical backing for the repairs to Theorem 6.2, including the direct Havel–Hakimi computation on the unique  $k = 0$  graph and the refutation of the enumeration claim that repair K2 replaced by a closed-form diameter argument.

## B.3 The budget chain

### `problems/wowii/w61_r4_low.py`

Certifies (7.1)–(7.3), the position-independence identity in the proof of Theorem 7.2, both clauses of Lemma 4.5, Lemma 7.1, and Theorem 4.6 as the case  $L = 0$ : 6 403 connected

graphs, 34040 reductio trajectory checks, every maximum independent set tested rather than one, canonical sort plus three adversarial tie-breaks. Emits the per- $L$  slack minima quoted in Remark 7.4 and the hard-core-frame  $L$  histogram.

**problems/wowii/w61\_r4\_slack.py**

The corpus deliberately enriched with near-split graphs ( $B$  a clique minus a few random edges), which is the regime Proposition 7.9 lives in: 5 913 graphs, 26 514 checks, 1 240 instances with  $L = 1$ , in all of which  $(\deg_A(b_0), \bar{m}) = (1, 0)$ .

**problems/wowii/w61\_r4\_thmSL.py**

Certifies *every intermediate assertion* of the proof of Theorem 7.3 separately, not only its conclusion: the eleven labelled steps, over 22 450 trajectories with five tie-breaks each. Produces the  $s_j$  histogram of Remark 7.4 and the 1 925 counterexamples to the stronger statement the theorem is *not*.

**problems/wowii/w61\_r4\_mb.py**

Numerical backing for Theorem 7.6 and Propositions 7.9, 7.11.

**problems/wowii/w61\_r5\_generalS.py**

Closes the general- $s$  species: checks each toolkit lemma off the reductio hypothesis, in the regime where a reductio-only statement would be invalid.

## B.4 The fan family and the rigidity chain

**problems/wowii/w61\_r4\_fanL.py**

Enumerates  $\text{Fan}(\tau, L \geq 2)$  degree sequences over a parameter box and reports  $\alpha$  – residue for each: 63 239 strict and 99 619 superset sequences, 0 with residue  $= \alpha$ , and the control  $L = 1$  with 106 811 sequences and 0 survivors. An enlarged box (**w61\_r4\_fanL\_big**) reaches 1 098 141 strict and 1 700 094 superset sequences; this is the source of Observation 8.10.

**problems/wowii/w61\_r4\_fanmech.py**

The mechanism check of Section 1.3’s firewall: 64 911 labelled Havel–Hakimi runs under canonical and randomised tie-breaks, testing which entries occupy which blocks. Includes a counterfactual-availability control — the same test applied to shapes breaking exactly one hypothesis of Definition 8.1 returns residue  $= \alpha$  in 36 650 of 46 138  $B$ -clique cases, so the test discriminates and is not vacuously negative.

**problems/wowii/w61\_r4\_fanres.py, w61\_r4\_fanE.py, w61\_r4\_fan.py**

Certify the residue-mass count of Lemma 8.4, the five structural hypotheses of Lemma 8.8 (508 239 high-phase runs, three tie-breaks each) and the escape budget of Lemma 8.6; report the residue partition in 450 303 superset sequences.

**problems/wowii/w61\_r5\_bcheck.py**

Independent confirmation of Observation 8.5’s second half for  $L = 2, \dots, 9$  against direct simulation.

**problems/wowii/w61\_r5\_rig.py**

Certifies Lemma 9.1, Corollary 9.2 and all five conclusions of Theorem 9.3, with its own labelled residue implementation, its own graph code and its own corpora, importing no earlier script. Contains the targeted frame generator of Remark 9.6 and the  $(L, \nu)$  histogram that measures the scope demarcation of Remark 9.4.

**problems/wowii/w61\_r5\_gfan2.py**

Certifies Theorem 10.4’s three trajectories and its eight-row  $L = 3$  table, and runs the  $\nu \leq 6$

enumeration through the three functions `tail_check` / `escape_check` / `complete_check`.

`problems/wowii/w61_r12_gfannu_embed.py`

The instrumented re-run that produced Appendix C: the same check as the previous entry, re-instrumented so that every intermediate column is printed rather than summarised, together with the zero-padding inertness control and the closed-form evaluation of  $S(\nu)$  independent of the enumeration loop.

`problems/wowii/w61_r13_padding_ext.py`

Extends the padding-inertness control to full scope: 964 rows with  $E = 0$  and all 1745 enumerated  $E \geq 1$  shapes for  $\nu \leq 6$ , times seven paddings, giving 18 963 (list, padding) pairs with 0 padding-dependent step counts.

`problems/wowii/w61_r12_gfannu_ext.py`

The  $\nu \leq 10$  extension of Remark 12.3, to the same specification.

`problems/wowii/w61_r12_c1_probe.py`, `w61_r12_c1k2.py`

Probe and certificate for Conjecture 12.1: the first evaluates  $s_0(\lambda)$  over every partition of every relevant weight in range; the second certifies the two-part case over all 10 268 terminating instances with  $n \leq 28$ .

## B.5 Adjudication scripts

Each review round is reproduced from scratch before any of its findings is upheld. These scripts are written to be maximally hostile to the paper: where this text prints data they parse the source file and diff it against their own recomputation, so that a transcription slip surfaces rather than being laundered.

`problems/wowii/w61_adjudicate.py`, `w61_adjudicate_r4.py`, `w61_r4_a2check.py`,  
`w61_r5_a2check.py`, `w61_r6_q14check.py`

Reproductions of the rounds behind rows R-1–R-13, each with an independent labelled Havel–Hakimi implementation.

`problems/wowii/w61_r10_adjudicate.py`, `w61_r11_adjudicate.py`, `w61_r12_adjudicate.py`,  
`w61_r13_adjudicate.py`

Reproductions of the four later rounds: respectively the budget chain, Lemma 8.12, the second-family pass on Sections 7–8, and the refuting round of Section 11.2. The last is the one described in Remark 11.5; it reports `FAILURE_COUNT = 0`.

`problems/wowii/w61_S3_76_sol_check.py`, `w61_S3_GFAN_sol_check.py`, `w61_S3_TAIL_spark_check.py`,  
`w61_S3B_K_hh.py`, `w61_S3B_K_exh.py`, `w61_S3B_K_avail.py`, `w61_S3B_K_probe.py`,  
`w61_S3B_T3_probe.py`

The reviewing sessions’ *own* checkers, archived as they were received. They were re-run independently during adjudication; one of them — the round-B judge’s exhaustive enumerator — performed 5 842 009 trajectories under exhaustive tie-break enumeration over 28 163 graphs.

`problems/wowii/w61_r10_build_spark_brief.py`, `w61_r10_build_sol_brief.py`,  
`w61_r12_build_gfannu_brief.py`, `w61_r12_build_sol_gfan_brief.py`

The brief builders. Each performs a leak check before emitting, verifying that the brief contains no expected answer, no pre-declared failure mode and no other reviewer’s report. The defect described in Section 10 is an omission in the last of these — four statements cited by name and not pasted — and repairing it is a change to that script’s appendix block.

**C** The  $\nu \leq 6$  enumeration, in full

This appendix is what makes Theorem 11.1 reproducible from this text rather than from an archived file. Its predecessor cited the enumeration by filename, and a reviewing session correctly refused to certify a column it could not recompute; everything below is the repair.

### C.1 The object being enumerated, specified exactly

At the start of step  $p+1$  of a  $\text{GFan}(\tau, L, \nu)$  run under the reductio hypothesis, the Havel–Hakimi value list is, by Lemma 10.1, exactly: a  $C$ -part of  $L+1$  entries, the  $c$ -th equal to  $L+e_c$  with  $e_c \geq 0$  and  $\sum_c e_c = E$ ; an  $A'$ -part which is a partition  $\lambda$  of  $2\nu - E$ ; and zero entries. Five conventions fix the enumeration.

1. *Unlabelled.* The step count of a Havel–Hakimi list depends only on the value multiset, so the  $C$ -part is enumerated as a multiset  $\{L + e_c\}$  — i.e.  $e$  is a partition of  $E$  into at most  $L + 1$  non-negative parts — and the  $A'$ -part as a partition of  $2\nu - E$ . Distinct labellings of the same multiset are the same row.
2. *Zero entries are inert*, verified rather than assumed. The block at each step is the  $D$  largest non-head entries, so a zero enters the block only when fewer than  $D$  positive entries remain, and in that case the run aborts either way; the zero padding in the scripts is therefore a convenience and  $|A'|$  never has to be pinned down. As a control, all 964 rows with  $E = 0$  and all 1745 enumerated  $E \geq 1$  shapes for  $\nu \leq 6$  were padded with 0, 1, 2, 3, 4, 8, 13 extra zeros — 18963 (list, padding) pairs — with 0 padding-dependent step counts.
3. *No graphicality filter is applied, and none is needed.* The enumeration is a superset argument: it ranges over every multiset the reductio hypothesis *could* produce and shows that none of them clears in exactly  $L$  steps except rows that Lemma 10.3 independently forbids. Adding a graphicality or feasibility filter can only remove rows, so the conclusion is monotone in the right direction.
4. *“Clears in exactly  $L$  steps”* means: iterating *head-deletes-the- $D$ -largest* from the list above reaches all-zeros in exactly  $L$  deletions. A run that would drive a zero entry negative, or whose head exceeds the number of remaining entries, returns “not a step sequence” and is not a survivor.
5. *Parameter ranges.*  $L \geq \nu + 1$  by Corollary 7.8;  $E \geq 1 \Rightarrow L \leq 2\nu - E$  by Lemma 10.2, which together force  $E \leq \nu - 1$ ; and the  $E = 0$  rows split at  $\lambda_1$  into the range  $L \geq \lambda_1$  where Lemma 8.12 applies and the finitely many boundary rows  $\nu + 1 < L < \lambda_1 < 2\nu$ .

### C.2 The $E = 0$ column, printed in full

By Lemma 8.12, for  $L \geq \lambda_1$  a row clears in exactly  $L$  steps if and only if  $s_0(\lambda) = \lambda_1$ , where  $s_0(\lambda) = \text{steps}([\lambda]_{\lambda_1+1} \cup \lambda)$  — a quantity of  $\lambda$  alone, computable by hand with one Havel–Hakimi run. Below, every partition of  $2\nu$  is listed as  $\lambda : s_0(\lambda)$ , survivors ( $s_0 = \lambda_1$ ) in bold. The counts are  $p(2\nu)$ , so the lists are complete by inspection.

$$\nu = 1 - p(2) = 2: \quad \mathbf{2:2} \cdot 1+1:2$$

$$\nu = \mathbf{2} - p(4) = 5: \quad \mathbf{4:4} \cdot \mathbf{3+1:4} \cdot \mathbf{2+2:3} \cdot \mathbf{2+1+1:3} \cdot \mathbf{1+1+1+1:3}$$

$$\nu = \mathbf{3} - p(6) = 11: \quad \mathbf{6:6} \cdot \mathbf{5+1:6} \cdot \mathbf{4+2:5} \cdot \mathbf{4+1+1:5} \cdot \mathbf{3+3:4} \cdot \mathbf{3+2+1:4} \cdot \mathbf{3+1+1+1:5} \cdot \mathbf{2+2+2:4} \cdot \mathbf{2+2+1+1:4} \cdot \mathbf{2+1+1+1+1:4} \cdot \mathbf{1+1+1+1+1+1:4}$$

$$\nu = 4 \text{ --- } p(8) = 22: \quad \mathbf{8:8} \cdot 7+1:8 \cdot 6+2:7 \cdot 6+1+1:7 \cdot 5+3:6 \cdot 5+2+1:6 \cdot 5+1+1+1:7 \cdot 4+4:5$$

$3+2+1+1+1:5 \cdot 3+1+1+1+1+1:6 \cdot 2+2+2+2:4 \cdot 2+2+2+1+1:5 \cdot 2+2+1+1+1+1:5 \cdot 2+1+1+1+1+1+1:5$   
 $\cdot 1+1+1+1+1+1+1:5$   
 $\nu = 5 \text{ — } p(10) = 42: \mathbf{10:10} \cdot 9+1:10 \cdot 8+2:9 \cdot 8+1+1:9 \cdot 7+3:8 \cdot 7+2+1:8 \cdot 7+1+1+1:9$   
 $\cdot 6+4:7 \cdot 6+3+1:7 \cdot 6+2+2:8 \cdot 6+2+1+1:8 \cdot 6+1+1+1+1:8 \cdot 5+5:6 \cdot 5+4+1:6 \cdot 5+3+2:7 \cdot$   
 $5+3+1+1:7 \cdot 5+2+2+1:7 \cdot 5+2+1+1+1:7 \cdot 5+1+1+1+1+1:8 \cdot 4+4+2:6 \cdot 4+4+1+1:6 \cdot 4+3+3:6 \cdot$   
 $4+3+2+1:6 \cdot 4+3+1+1+1:6 \cdot 4+2+2+2:6 \cdot 4+2+2+1+1:7 \cdot 4+2+1+1+1+1:7 \cdot 4+1+1+1+1+1+1:7$   
 $\cdot 3+3+3+1:5 \cdot 3+3+2+2:5 \cdot 3+3+2+1+1:6 \cdot 3+3+1+1+1+1:6 \cdot 3+2+2+2+1:6 \cdot 3+2+2+1+1+1:6 \cdot$   
 $3+2+1+1+1+1+1:6 \cdot 3+1+1+1+1+1+1+1:7 \cdot 2+2+2+2+2:5 \cdot 2+2+2+2+1+1:5 \cdot 2+2+2+1+1+1+1:6$   
 $\cdot 2+2+1+1+1+1+1+1:6 \cdot 2+1+1+1+1+1+1+1+1:6 \cdot 1+1+1+1+1+1+1+1+1+1:6$   
 $\nu = 6 \text{ — } p(12) = 77: \mathbf{12:12} \cdot 11+1:12 \cdot 10+2:11 \cdot 10+1+1:11 \cdot 9+3:10 \cdot 9+2+1:10 \cdot$   
 $9+1+1+1:11 \cdot 8+4:9 \cdot 8+3+1:9 \cdot 8+2+2:10 \cdot 8+2+1+1:10 \cdot 8+1+1+1+1:10 \cdot 7+5:8 \cdot 7+4+1:8$   
 $\cdot 7+3+2:9 \cdot 7+3+1+1:9 \cdot 7+2+2+1:9 \cdot 7+2+1+1+1:9 \cdot 7+1+1+1+1+1:10 \cdot 6+6:7 \cdot 6+5+1:7$   
 $\cdot 6+4+2:8 \cdot 6+4+1+1:8 \cdot 6+3+3:8 \cdot 6+3+2+1:8 \cdot 6+3+1+1+1:8 \cdot 6+2+2+2:8 \cdot 6+2+2+1+1:9$   
 $\cdot 6+2+1+1+1+1:9 \cdot 6+1+1+1+1+1+1:9 \cdot 5+5+2:7 \cdot 5+5+1+1:7 \cdot 5+4+3:7 \cdot 5+4+2+1:7 \cdot$   
 $5+4+1+1+1:7 \cdot 5+3+3+1:7 \cdot 5+3+2+2:7 \cdot 5+3+2+1+1:8 \cdot 5+3+1+1+1+1:8 \cdot 5+2+2+2+1:8$   
 $\cdot 5+2+2+1+1+1:8 \cdot 5+2+1+1+1+1+1:8 \cdot 5+1+1+1+1+1+1+1:9 \cdot 4+4+4:6 \cdot 4+4+3+1:6 \cdot$   
 $4+4+2+2:6 \cdot 4+4+2+1+1:7 \cdot 4+4+1+1+1+1:7 \cdot 4+3+3+2:6 \cdot 4+3+3+1+1:7 \cdot 4+3+2+2+1:7 \cdot$   
 $4+3+2+1+1+1:7 \cdot 4+3+1+1+1+1+1:7 \cdot 4+2+2+2+2:7 \cdot 4+2+2+2+1+1:7 \cdot 4+2+2+1+1+1+1:8$   
 $\cdot 4+2+1+1+1+1+1+1:8 \cdot 4+1+1+1+1+1+1+1+1:8 \cdot 3+3+3+3:6 \cdot 3+3+3+2+1:6 \cdot 3+3+3+1+1+1:6 \cdot$   
 $3+3+2+2+2:6 \cdot 3+3+2+2+1+1:6 \cdot 3+3+2+1+1+1+1:7 \cdot 3+3+1+1+1+1+1+1:7 \cdot 3+2+2+2+2+1:6$   
 $\cdot 3+2+2+2+1+1+1:7 \cdot 3+2+2+1+1+1+1+1+1:7 \cdot 3+2+1+1+1+1+1+1+1+1:7 \cdot 3+1+1+1+1+1+1+1+1+1:8$   
 $\cdot 2+2+2+2+2+2:6 \cdot 2+2+2+2+2+1+1:6 \cdot 2+2+2+2+1+1+1+1:6 \cdot 2+2+2+1+1+1+1+1+1+1:7$   
 $\cdot 2+2+1+1+1+1+1+1+1+1+1:7 \cdot 2+1+1+1+1+1+1+1+1+1+1+1:7 \cdot 1+1+1+1+1+1+1+1+1+1+1+1:7$

**Reading.** For every  $\nu \leq 6$  the only partition of  $2\nu$  with  $s_0(\lambda) = \lambda_1$  is the single part  $[2\nu]$ . And  $[2\nu]$  has unique maximum  $2\nu \geq 2$  with second-largest entry  $0 \leq 2\nu - 2$ , so Lemma 10.3 kills it. Hence the rows with  $E = 0$  and  $L \geq \lambda_1$  contribute nothing, for every  $\nu \leq 6$ .

### C.3 The $E = 0$ boundary rows, printed in full

Outside Lemma 8.12's range the rows  $\nu + 1 \leq L < \lambda_1$  are checked directly. They are few, and every survivor is listed with its Lemma 10.3 certificate, written as the pair (unique maximum  $w$ , second-largest entry).

| $\nu$ | pairs $(L, \lambda)$ checked | survivors                                   | certificates $(w, 2\text{nd})$ |
|-------|------------------------------|---------------------------------------------|--------------------------------|
| 1     | 0                            | none                                        | —                              |
| 2     | 1                            | $(3, [4])$                                  | $(4, 0)$                       |
| 3     | 3                            | $(5, [6])$                                  | $(6, 0)$                       |
| 4     | 7                            | $(5, [7, 1]), (7, [8])$                     | $(7, 1), (8, 0)$               |
| 5     | 14                           | $(7, [9, 1]), (9, [10])$                    | $(9, 1), (10, 0)$              |
| 6     | 26                           | $(7, [10, 1, 1]), (9, [11, 1]), (11, [12])$ | $(10, 1), (11, 1), (12, 0)$    |

Every certificate has a unique maximum  $w$  with second-largest entry  $\leq w - 2$ , so Lemma 10.3 kills every boundary survivor.

### C.4 The $E \geq 1$ rows: a closed-form count, and every survivor

First the count, so that the “shapes tested” column of Theorem 11.1 stops being an opaque number. For  $E \geq 1$ ,  $L$  ranges over  $\nu + 1 \leq L \leq 2\nu - E$  — that is  $\nu - E$  values, so  $E \leq \nu - 1$ ; the

escape multiset  $e$  is a partition of  $E$  into at most  $L + 1$  parts, and  $L + 1 \geq \nu + 2 > E$ , so all  $p(E)$  partitions of  $E$  occur; and  $\lambda$  is any partition of  $2\nu - E$ . Hence

$$S(\nu) = \sum_{E=1}^{\nu-1} (\nu - E) p(E) p(2\nu - E).$$

For  $\nu = 6$  this is  $5 \cdot 1 \cdot p(11) + 4 \cdot 2 \cdot p(10) + 3 \cdot 3 \cdot p(9) + 2 \cdot 5 \cdot p(8) + 1 \cdot 7 \cdot p(7) = 280 + 336 + 270 + 220 + 105 = 1211$ , with per- $E$  split  $\{1 : 280, 2 : 336, 3 : 270, 4 : 220, 5 : 105\}$  reproduced term by term by the instrumented run; the same formula gives 0, 3, 24, 110, 397, 1211 for  $\nu = 1, \dots, 6$ . A reviewing session derived this identity independently and evaluated it from its own partition table.

Of those  $S(\nu)$  shapes, the ones clearing in exactly  $L$  steps number 0, 1, 4, 9, 20, 38 for  $\nu = 1, \dots, 6$ . Here they are, each with its Lemma 10.3 certificate ( $w, 2nd$ );  $e$  lists the positive escape parts only.

$\nu = 2$  (1): L3 E1  $e=1$   $\lambda=3$  (3,0)

$\nu = 3$  (4): L4 E1  $e=1$   $\lambda=5$  (5,0) · L5 E1  $e=1$   $\lambda=5$  (5,0) · L4 E2  $e=1+1$   $\lambda=4$  (4,0) · L4 E2  $e=2$   $\lambda=3+1$  (3,1)

$\nu = 4$  (9): L6 E1  $e=1$   $\lambda=7$  (7,0) · L7 E1  $e=1$   $\lambda=7$  (7,0) · L5 E2  $e=1+1$   $\lambda=6$  (6,0) · L5 E2  $e=2$   $\lambda=5+1$  (5,1) · L6 E2  $e=1+1$   $\lambda=6$  (6,0) · L6 E2  $e=2$   $\lambda=5+1$  (5,1) · L5 E3  $e=1+1+1$   $\lambda=5$  (5,0) · L5 E3  $e=2+1$   $\lambda=4+1$  (4,1) · L5 E3  $e=3$   $\lambda=3+1+1$  (3,1)

$\nu = 5$  (20): L6 E1  $e=1$   $\lambda=8+1$  (8,1) · L8 E1  $e=1$   $\lambda=9$  (9,0) · L9 E1  $e=1$   $\lambda=9$  (9,0) · L6 E2  $e=2$   $\lambda=7+1$  (7,1) · L7 E2  $e=1+1$   $\lambda=8$  (8,0) · L7 E2  $e=2$   $\lambda=7+1$  (7,1) · L8 E2  $e=1+1$   $\lambda=8$  (8,0) · L8 E2  $e=2$   $\lambda=7+1$  (7,1) · L6 E3  $e=1+1+1$   $\lambda=7$  (7,0) · L6 E3  $e=2+1$   $\lambda=6+1$  (6,1) · L6 E3  $e=3$   $\lambda=5+1+1$  (5,1) · L7 E3  $e=1+1+1$   $\lambda=7$  (7,0) · L7 E3  $e=2+1$   $\lambda=6+1$  (6,1) · L7 E3  $e=3$   $\lambda=5+1+1$  (5,1) · L6 E4  $e=1+1+1+1$   $\lambda=6$  (6,0) · L6 E4  $e=2+1+1$   $\lambda=5+1$  (5,1) · L6 E4  $e=2+2$   $\lambda=4+2$  (4,2) · L6 E4  $e=2+2$   $\lambda=4+1+1$  (4,1) · L6 E4  $e=3+1$   $\lambda=4+1+1$  (4,1) · L6 E4  $e=4$   $\lambda=3+1+1+1$  (3,1)

$\nu = 6$  (38): L8 E1  $e=1$   $\lambda=10+1$  (10,1) · L10 E1  $e=1$   $\lambda=11$  (11,0) · L11 E1  $e=1$   $\lambda=11$  (11,0) · L7 E2  $e=1+1$   $\lambda=9+1$  (9,1) · L8 E2  $e=2$   $\lambda=9+1$  (9,1) · L9 E2  $e=1+1$   $\lambda=10$  (10,0) · L9 E2  $e=2$   $\lambda=9+1$  (9,1) · L10 E2  $e=1+1$   $\lambda=10$  (10,0) · L10 E2  $e=2$   $\lambda=9+1$  (9,1) · L7 E3  $e=2+1$   $\lambda=8+1$  (8,1) · L7 E3  $e=3$   $\lambda=7+1+1$  (7,1) · L8 E3  $e=1+1+1$   $\lambda=9$  (9,0) · L8 E3  $e=2+1$   $\lambda=8+1$  (8,1) · L8 E3  $e=3$   $\lambda=7+1+1$  (7,1) · L9 E3  $e=1+1+1$   $\lambda=9$  (9,0) · L9 E3  $e=2+1$   $\lambda=8+1$  (8,1) · L9 E3  $e=3$   $\lambda=7+1+1$  (7,1) · L7 E4  $e=1+1+1+1$   $\lambda=8$  (8,0) · L7 E4  $e=2+1+1$   $\lambda=7+1$  (7,1) · L7 E4  $e=2+2$   $\lambda=6+2$  (6,2) · L7 E4  $e=2+2$   $\lambda=6+1+1$  (6,1) · L7 E4  $e=3+1$   $\lambda=6+1+1$  (6,1) · L7 E4  $e=4$   $\lambda=5+1+1+1$  (5,1) · L8 E4  $e=1+1+1+1$   $\lambda=8$  (8,0) · L8 E4  $e=2+1+1$   $\lambda=7+1$  (7,1) · L8 E4  $e=2+2$   $\lambda=6+2$  (6,2) · L8 E4  $e=2+2$   $\lambda=6+1+1$  (6,1) · L8 E4  $e=3+1$   $\lambda=6+1+1$  (6,1) · L8 E4  $e=4$   $\lambda=5+1+1+1$  (5,1) · L7 E5  $e=1+1+1+1+1$   $\lambda=7$  (7,0) · L7 E5  $e=2+1+1+1$   $\lambda=6+1$  (6,1) · L7 E5  $e=2+2+1$   $\lambda=5+2$  (5,2) · L7 E5  $e=2+2+1$   $\lambda=5+1+1$  (5,1) · L7 E5  $e=3+1+1$   $\lambda=5+1+1$  (5,1) · L7 E5  $e=3+2$   $\lambda=4+2+1$  (4,2) · L7 E5  $e=3+2$   $\lambda=4+1+1+1$  (4,1) · L7 E5  $e=4+1$   $\lambda=4+1+1+1$  (4,1) · L7 E5  $e=5$   $\lambda=3+1+1+1+1$  (3,1)

**Reading.** In every one of these 72 rows the residue  $\lambda$  has a unique maximum  $w \geq 3$  whose second-largest entry is  $\leq w - 2$ : the pairs are  $(w, 0)$ , or  $(w, 1)$  with  $w \geq 3$ , or  $(w, 2)$  with  $w \geq 4$ . So Lemma 10.3 kills every  $E \geq 1$  survivor, for every  $\nu \leq 6$ . That is the “misses = none” column of Theorem 11.1, printed as data rather than asserted as an output.

## C.5 What a re-reader has to do

To reproduce Theorem 11.1 from this text it suffices to (i) recompute  $s_0(\lambda)$  for the 159 partitions of C.2 — one Havel–Hakimi run each, all by hand — and confirm that the bolded survivor is the only one at each  $\nu$ ; (ii) check the 51 boundary pairs of C.3; (iii) evaluate  $S(\nu)$  from the closed form of C.4 and confirm that the 72 printed survivors are the complete survivor set for those 1745 shapes; and (iv) apply the Lemma 10.3 certificate to each of the  $9 + 72 + 6$  surviving rows. Only step (iii)’s completeness claim still asks the reader to trust a machine run, and it is a claim about

an explicitly specified, closed-form-counted finite set rather than about an absent file. That is the exact location of the demarcation line of Remark 11.2.

## References

- [1] Google DeepMind and contributors, *formal-conjectures*: a corpus of open conjectures stated in Lean 4, <https://github.com/google-deepmind/formal-conjectures>. The statement treated here is `FormalConjectures/WrittenOnTheWallIII/GraphConjecture61.lean`, with supporting definitions in `FormalConjecturesForMathlib/Combinatorics/SimpleGraph/Residue.lean` and `.../Induced.lean`. The corpus attributes the conjecture to the *Written on the Wall II* collection generated by the `Graffiti.pc` program; we consumed the formal file and did not consult the original collection.
- [2] V. Havel, *A remark on the existence of finite graphs*, Časopis Pěst. Mat. **80** (1955), 477–480. Bibliographic details as supplied by our automated literature pass.
- [3] S. L. Hakimi, *On realizability of a set of integers as degrees of the vertices of a linear graph*, J. Soc. Indust. Appl. Math. **10** (1962), 496–506. Bibliographic details as supplied by our automated literature pass.
- [4] O. Favaron, M. Mahéo and J.-F. Saclé, *On the residue of a graph*, J. Graph Theory **15** (1991), 39–64. The source of Fact 2.2. Bibliographic details as supplied by our automated literature pass; the inequality itself is verified computationally on the whole exhaustive range but is not reproved here (Section 13).
- [5] J. R. Griggs and D. J. Kleitman, *Independence and the Havel–Hakimi residue*, Discrete Math. **127** (1994), 209–212. A short proof of Fact 2.2. Bibliographic details as supplied by our automated literature pass.
- [6] The mathlib Community, *The Lean mathematical library*, Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020), 367–381. <https://github.com/leanprover-community/mathlib4>. The source of the `SimpleGraph.diam` and degree-sequence definitions against which the residue transcription of Section 1.1 was checked.
- [7] H. Chen, *Machine-Verified Resolutions of Four OEIS Conjectures*, 2026. The companion report of the same pipeline on four resolved conjectures, each formalized in Lean 4; it describes the review protocol summarised in Section 1.3.
- [8] H. Chen, *The parity subgroup of  $S_m \times S_n$  is 2-generated: a proof of Fernandes’ conjecture*, 2026.
- [9] H. Chen, *Structure Theory of Finite 677 Magmas: a working paper*, 2026. The companion working paper on an unfinished campaign, from which the per-assertion verification-marker discipline of Section 1.3 is taken.
- [10] B. Romera-Paredes, M. Barekatin, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli and A. Fawzi, *Mathematical discoveries from program search with large language models*, Nature **625** (2024), 468–475.

- [11] T. F. Bloom (site), T. Tao (community database) and contributors, *Erdős Problems* and the associated AI-contributions wiki. <https://www.erdosproblems.com>, <https://github.com/teorth/erdosproblems>. Cited for its apparatus of caveats about selection bias and about the difference between an argument and a verified proof, which the tiering of Section 1.3 is intended to respect.